

Cash Wage Payments, Tax Evasion, and Informality*

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Abstract

This paper studies how restricting cash wage payments affects firms’ compliance and labor-market behavior. We study a nationwide reform in Uruguay that prohibited cash wage payments and required wages to be paid electronically, exploiting pre-reform sector-level cash intensity in a difference-in-differences design combining administrative and survey data. We find that the reform sharply reduced cash wage payments, lowered formal firm activity and payroll employment, and increased reported wages. Survey evidence shows less wage underreporting and more informal employment. Overall, the contraction in formal activity more than offsets the increase in reported wages, leading to a decline in reported payroll.

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1. Introduction

Governments increasingly rely on third-party information and electronic records to improve tax enforcement. Making transactions more observable can limit firms’ ability to conceal economic activity, motivating efforts to reduce the use of cash, whose low traceability can facilitate tax evasion and other illicit activities (e.g., Rogoff, 2017; Gupta et al., 2017). Yet firms can often evade taxes along multiple margins: when enforcement makes one form of noncompliance more difficult, firms may comply or shift activity toward forms that remain outside the tax system. Restrictions on cash can therefore improve reporting among formal firms while pushing some economic activity fully out of the formal system.

Labor markets provide a natural setting in which to study this tradeoff. Employers can evade payroll taxes by underreporting workers’ earnings while remaining formal, or by employing workers entirely off the books. Cash can facilitate both forms of noncompliance because it leaves little third-party record, and recent evidence finds that cash is the primary method used to pay formal workers off the books (Feinmann et al., 2024). Thus, requiring wages to be paid electronically could, in principle, reduce underreporting, but may also increase the cost of formal operation for firms that relied on cash to conceal part of their wage bill. Better reporting among formal firms may thus be offset by a shift toward informality, making the effect on the tax base ambiguous.

In this paper, we study these margins using a reform in Uruguay that prohibited cash wage payments and required wages to be disbursed electronically. The reform was part of a broader effort to increase financial inclusion and formalization in a country where account ownership was relatively low for its income level (Demirgüç-Kunt et al., 2022).¹ We combine administrative payroll records covering the universe of formally registered firms with nationally representative household survey data on payment methods, informality, and wage underreporting. Our difference-in-differences strategy exploits cross-sector variation in the share of workers paid in cash before the reform, with more cash-intensive sectors more exposed to the mandate. We first show that the mandate substantially reduced the use of cash for wage payments. Each percentage point of greater pre-reform cash intensity is associated with a 0.66-percentage-point larger decline in cash wage payments among formal workers, closing roughly two-thirds of the initial gap between high- and low-cash-intensity sectors. The decline in cash payments is mirrored almost one-for-one by an increase in bank transfers.

¹ The reform also incorporated other measures to boost financial inclusion, such as VAT rebates for debit card purchases and a mandate to disburse pensions and other social assistance programs with electronic methods, among others. We describe the institutional context in more detail in Section 2. See the official government website for a discussion of the several goals of the reform, <https://inclusionfinanciera.uy/por-que/>, accessed in April 2024.

The reform also changed firms’ formal labor-market activity. Firms in sectors with 50 percentage points greater pre-reform cash intensity are about 7.8 percentage points less likely to remain formally active in the long run. Among firms that remain active throughout the sample, formal employment falls by about 5.7 log points, while average reported wages increase by about 4.6 log points. The decline in formal activity and the increase in reported wages are concentrated among smaller firms, which are more likely to operate near the margin of informality and to engage in wage underreporting (Ulyssea, 2018; Feinmann et al., 2024).²

Evidence from household survey data points to informality and underreporting of formal earnings as key margins of adjustment. We find that formally employed workers in more cash-intensive sectors show no significant increase in reported earnings, while becoming significantly less likely to underreport their earnings for tax purposes. At the same time, workers in these sectors become more likely to be informally employed. For a 50-percentage-point difference in pre-reform cash intensity, underreporting falls by about 1.8 percentage points and informality rises by about 5.1 percentage points. We do not detect a statistically significant effect on employment among survey respondents who can be assigned a current or previous sector. We also find no detectable change in hours worked in either the administrative or survey data. Together, these results suggest that the reform reduced underreporting among formal workers while inducing some activity to move into informality.

Finally, these responses have important implications for tax revenue. A direct measure of firm payroll, which combines changes in formal activity, employment, and reported wages, declines substantially in more cash-intensive sectors. For firms in sectors with 50 percentage points greater pre-reform cash intensity, our estimates imply a long-run decline equivalent to 13.4 percent of baseline payroll. Thus, the gains from improved wage reporting among firms that remain formal can be more than offset by reductions in formal firms and jobs.

Our paper contributes first to the literature on tax enforcement and third-party information. A large literature studies how information and reporting requirements affect tax compliance (e.g., Slemrod and Yitzhaki, 2002; Andreoni et al., 1998). Third-party reporting can sharply limit evasion (Kleven et al., 2011; Pomeranz, 2015; Naritomi, 2019), although firms may respond by shifting evasion toward margins that remain less observable (Carrillo et al., 2017; Slemrod et al., 2017). Our findings highlight a different margin of adjustment. When firms can choose whether to participate in the formal sector, restrictions on cash payments can reduce evasion among activity that remains formal while shifting some activity outside the formal system altogether. More broadly, our results relate to work on tax capacity in developing countries, where limited enforcement capacity and informality shape both tax design and compliance (Besley and Persson, 2013; Best et al., 2015; Gordon and Li, 2009).

² Small firms are also more likely to pay their workers in cash; see Appendix B.

Second, we contribute to the growing literature on cash, electronic payments, and tax compliance. Recent work studies how policies promoting electronic payments affect savings and financial behavior (Bachas et al., 2021), the adoption of payment technologies (Higgins, 2024; Crouzet et al., 2023), and tax compliance. Brockmeyer and Sáenz Somarriba (2025) study VAT rebates for debit-card purchases in Uruguay and find little effect on tax reporting, while Das et al. (2023) find that India’s demonetization increased firms’ reported sales. We study a different margin through which payment technology can affect compliance: restrictions on cash wage payments, which may increase the observability of labor compensation. Our results show that restricting cash wage payments can reduce underreporting among formal workers, while shifting some economic activity into full informality. Thus, the effects of reducing cash use depend not only on whether electronic payments improve reporting among observed firms, but also on whether firms and workers remain within the formal reporting system.

Finally, our paper relates to the literature on informality and the underreporting of labor earnings. A large literature studies the prevalence and consequences of informal employment and the role of taxes and registration costs in firms’ formalization decisions (Williams and Lansky, 2013; Bitran, 2014; La Porta and Shleifer, 2014; Ulyssea, 2018, 2020). A related literature shows that substantial evasion can remain even within formal employment. Kumler et al. (2020) infer sizable underreporting of labor earnings in Mexico, while Bíró et al. (2022) show that collusive underreporting contributes to minimum-wage bunching in Hungary. Lauletta and Bérgho (2025) show that strengthening the link between reported earnings and future pension benefits increases reported earnings in Uruguay, highlighting the role of worker incentives in wage reporting. Feinmann et al. (2024) document widespread employer–employee collusion in Brazil, with cash payments commonly used to disburse earnings off the books. Our findings connect these forms of noncompliance by showing that restricting cash wage payments reduces underreporting within formal employment, while the accompanying rise in informality suggests that policies targeting one form of evasion can induce substitution toward another.

The rest of the paper is structured as follows. Section 2 describes the context and institutional setting. Section 3 introduces the conceptual framework. Section 4 describes the data and the econometric strategy. Section 5 presents the results. Section 6 discusses the revenue implications of our findings. Section 7 concludes.

2. Context

The Uruguayan context Uruguay has a well-established contributory social security system administered by the *Banco de Previsión Social* (BPS). Employers must register formal employees and report their earnings monthly, which determines contributions financing pensions, unemployment insurance, health insurance, workers’ compensation, disability benefits, and parental leave.

Despite relatively strong institutions, tax evasion and labor informality remain important. The tax administration estimates that approximately 20 percent of potential VAT revenue was evaded in 2012, equivalent to 2.5 percent of GDP ([Dirección General Impositiva, 2019](#)). During our 2014–2019 sample period, approximately 26 percent of employees were not registered with social security. Financial inclusion was also limited before the reform: in 2011, fewer than 25 percent of adults held a financial account, substantially below the rate in countries with comparable income levels (see panel (a) of Appendix Figure [B.XI](#)). Account ownership subsequently increased sharply following the reform (see panel (b) of Appendix Figure [B.XI](#)).

The reform In 2014, Uruguay adopted a broad financial inclusion reform intended to expand access to financial services, promote electronic payments, and reduce tax evasion and informality. The law included VAT rebates for electronic purchases, studied by [Brockmeyer and Sáenz Somarriba \(2025\)](#), and created fee-free “wage accounts” with no minimum-balance requirement.

The provision we study prohibited cash wage payments and required salaries to be disbursed through bank transfers, prepaid cards, mobile-payment applications, or other electronic instruments.³ Beginning in October 2015, workers could coordinate with employers to select an electronic payment method. Electronic wage payments became mandatory in May 2017.⁴ We refer to October 2015–April 2017 as the *phase-in period* and the period beginning in May 2017 as the *mandate period*.

The mandate was enforced through the existing labor-regulation system, including establishment inspections and worker complaints. Employers paying wages in cash could be fined the greater of 25 percent of the amount paid or twice the minimum wage. Because enforcement is imperfect, we first empirically estimate whether the reform actually reduced cash wage payments, and then examine its effects on formal employment, reported earnings, and informality.

³ The reform also required electronic disbursement of pensions and certain social benefits.

⁴ Exceptions and transitional arrangements applied to certain workers in rural areas and small localities, while domestic workers followed a separate implementation timetable.

3. Conceptual framework

We develop a simple static framework combining the canonical tax-evasion model of Allingham and Sandmo (1972) with the formal–informal firm choice in Ulyssea (2018). The framework illustrates how restricting cash wage payments affects both payroll-tax evasion within formal firms and selection into informality. Detailed assumptions and derivations are provided in Appendix F.

A continuum of risk-neutral firms draws productivity θ and produces output according to $y = \theta f(L)$, where $f' > 0$ and $f'' < 0$. Firms choose whether to operate formally or informally. A formal firm incurs a fixed cost F , pays payroll taxes at rate τ on the share $\gamma \in [0, 1]$ of its wage bill that it reports, and faces an expected penalty on the unreported share. Its profits are

$$\pi_F(\theta; L, \gamma, \eta) = \theta f(L) - [1 + \tau\gamma + \phi\tau(1 - \gamma)p(\gamma, \eta, L)] wL - F, \quad (1)$$

where w is an exogenous wage fixed across regimes, with reported wages equal to γw . The probability of detection is $p(\gamma, \eta, L)$, and $\phi > 1$ is the payment per unit of evaded tax upon detection, including repayment and the penalty. Enforcement, denoted by η , affects the probability of detection, which decreases with the reported share γ and increases with enforcement and firm size. We interpret the reform as raising η by increasing the expected cost of concealing wages within formal employment, potentially through greater payment traceability.

An informal firm pays neither registration costs nor payroll taxes, but faces increasing costs of operating at larger scale:

$$\pi_I(\theta; L) = \theta f(L) - w\kappa(L), \quad \kappa' > 0, \quad \kappa'' > 0. \quad (2)$$

These costs capture the greater detection risk faced by larger informal firms, as well as limited access to credit and contract enforcement.

Let (L_F^*, γ^*) and L_I^* denote the optimal choices under formality and informality. Under the joint-concavity, interiority, and comparative-static conditions discussed in Appendix F, the model implies:

$$\frac{\partial \gamma^*}{\partial \theta} > 0 \quad \text{and} \quad \frac{\partial \gamma^*}{\partial \eta} > 0. \quad (3)$$

More productive formal firms report a larger share of wages because their greater scale raises the probability of detection. Stronger enforcement similarly increases wage reporting by raising the expected cost of evasion.

A firm operates formally whenever

$$\pi_F(\theta; L_F^*, \gamma^*, \eta) \geq \pi_I(\theta; L_I^*). \quad (4)$$

Under the conditions stated in Appendix F, this defines a unique productivity cutoff $\theta^*(\eta)$: firms below the cutoff operate informally, while those above it operate formally. The cash wage ban raises the expected cost of operating formally while underreporting wages and therefore shifts this cutoff upward:

$$\frac{\partial \theta^*(\eta)}{\partial \eta} > 0. \quad (5)$$

Consequently, some relatively low-productivity formal firms move into informality, while firms that remain formal report a larger share of their wage bill.

Under these conditions, the framework highlights three empirical implications. First, reported wages should increase and underreporting should decline within continuing formal firms. Second, formal firm participation falls through the exit of relatively low-productivity firms, generating an extensive-margin decline in formal employment. Aggregate formal employment falls if employment changes at continuing formal firms do not offset this decline. Third, the effect on payroll-tax revenue is ambiguous. Writing revenue as

$$R(\eta) = \tau E_F(\eta) \tilde{w}_F(\eta), \quad (6)$$

where $E_F(\eta)$ is formal employment and $\tilde{w}_F(\eta)$ is the reported tax base per formal worker, improved reporting tends to raise revenue, while exit from formality tends to reduce it. The net effect also depends on employment adjustments and changes in the composition of formal employment.

4. Data and empirical strategy

4.1 Data

Administrative records Our primary data consist of monthly payroll records from the Social Security Administration, which cover the universe of formally registered firms from 2014 to 2019. For each firm, we observe its employees and their reported wages and monthly hours worked, as well as the firm's 2-digit ISIC sector and legal status. We construct a quarterly panel using observations for March, June, September, and December of each year. Employment, wages, hours, and payroll therefore refer to the quarter-end month rather than to totals or averages across all three months. Administrative wages and payroll are deflated

using the national consumer price index and expressed in September 2015 pesos.

We define a firm as active in a quarter if it reports at least one dependent employee in the quarter-end month. For active firm-quarters, we calculate payroll employment, average reported wages, and average monthly hours worked per employee. We restrict the sample to firms observed as active before the reform, from 2014Q1 through 2015Q3, and employing at least three workers on average across their observed active quarters during that period.⁵ The resulting data form a balanced quarterly panel from 2014Q1 through 2019Q4, including zero activity and payroll in inactive quarters. Baseline regressions for log employment, log average wages, and average hours worked further restrict the sample to firms active in all 24 quarters; activity and relative-payroll regressions retain firms that subsequently become inactive. Panel A of Table A.I reports summary statistics.

Household survey Our second data source is the nationally representative Continuous Household Survey (*Encuesta Continua de Hogares*, ECH), a quarterly repeated cross-section containing information on employment, earnings, weekly hours worked, social-security coverage, and industry of employment.

The ECH asks workers whether they receive their wages in cash, by bank transfer, prepaid card, or another method.⁶ We use these responses to construct indicators for cash and electronic wage payments. We also use the survey to measure each 2-digit ISIC sector’s pre-reform cash intensity, defined as the weighted share of workers paid in cash between 2014Q1 and 2015Q2, and merge this measure with the corresponding sector in the administrative records.

Because some sector-level means are estimated from relatively few observations, we apply empirical-Bayes shrinkage within broad industries. The procedure combines each sector’s weighted cash-payment mean and sampling variance in a random-effects model, pulling imprecisely estimated sectors toward their industry mean while leaving precisely estimated sectors largely unchanged (e.g., Fay III and Herriot, 1979; DerSimonian and Laird, 1986; Gelman and Hill, 2007). Similar shrinkage procedures are commonly used when estimated unit-level measures serve as inputs in subsequent analysis (e.g., Chetty et al., 2014a,b; Chetty and Hendren, 2018; Finkelstein et al., 2021; Abaluck et al., 2021). Results are similar using the unshrunk means.

We define a worker as formal if they contribute to a pension system through their main

⁵ Both datasets exclude public administration and defense and households as employers of domestic personnel.

⁶ The ECH groups cash and check payments together, which we refer to as cash payments. Independent evidence from the 2014 Global Findex indicates that cash-only payments accounted for approximately 94% of private-sector wage recipients not receiving wages through an account, card, or mobile phone (authors’ calculations using the World Bank’s [Global Findex Database](#)).

occupation, following standard practice in the literature (e.g., Tornarolli et al., 2014). The informality analysis includes both salaried and self-employed workers. The ECH also asks formal employees whether social-security contributions are calculated using all or only part of their earnings. We define underreporting as an indicator equal to 1 when only part of the worker’s earnings is declared, a measure designed to capture collusive evasion between employers and workers (Bergolo and Cruces, 2014).

The final ECH sample covers 2014Q1 through 2019Q4. The analysis is restricted to respondents aged 18 or older and, except in the employment regression, excludes public-sector workers. All baseline estimates use sampling weights.⁷ The employment regression includes respondents with an observed current sector or, when the current sector is unavailable, a reported sector from their previous job. Panel B of Table A.I presents summary statistics.⁸

4.2 Econometric strategy

We leverage variation in pre-reform cash intensity across sectors as a measure of differential exposure to the mandate, implementing a difference-in-differences design with continuous treatment intensity. Intuitively, sectors that relied minimally on cash for wage disbursements should be less affected than sectors where cash was the dominant payment method. Panel (a) of Figure 1 shows substantial variation in this measure across sectors.⁹ Specifically, we estimate the following equation:

$$Y_{ist} = \sum_{k \neq -1} \beta_k \mathbf{1}\{t = e + k\} \times \text{PreCashUse}_s + \alpha_i + \mu_{I(s)t} + \varepsilon_{ist}, \quad (7)$$

where Y_{ist} denotes an outcome for firm i in sector s and quarter t , and PreCashUse_s is the sector’s pre-reform share of workers paid in cash. We include firm fixed effects, α_i , and industry-by-quarter fixed effects, $\mu_{I(s)t}$, so that identification comes from differences in cash intensity across sectors within the same broad industry. Sectors are defined at the 2-digit ISIC level, while industries follow ISIC section groupings. The reform begins in 2015Q4, denoted by e , and the omitted quarter is 2015Q3.

The β_k coefficients measure how outcomes evolve in more cash-intensive sectors relative to less cash-intensive sectors. Our identifying assumption is that, absent the reform, these outcomes would have followed parallel trends within industries. The pre-reform coefficients

⁷ We report unweighted versions in Appendix Table A.II.

⁸ We exclude the bottom 10% of 2-digit ISIC sectors by pre-reform survey sample size, with 73 sectors represented in the baseline regressions after the remaining sample restrictions. Results are robust to retaining the excluded sectors.

⁹ Appendix Figure B.V shows the distribution after removing broad-industry fixed effects. Figure B.IV reports cash intensity for each sector within industry.

allow us to assess this assumption, while the post-reform coefficients trace the effects during the phase-in and mandate periods. Standard errors are clustered at the sector level, where exposure varies.

We apply the same approach to the household survey, replacing firm fixed effects with sector fixed effects and controlling for gender, age, education, and region-specific time shocks.¹⁰ Table 1 summarizes the long-run effects by pooling the final three quarters of the sample (2019Q2–2019Q4), relative to the three quarters immediately preceding the reform (2015Q1–2015Q3).

5. Results

5.1 Main results

Cash usage for wage payments We begin by exploring whether the reform effectively reduced the use of cash in wage payments. Panel (c) of Figure 1 reports estimates of equation (7) for the probability that a formal worker receives wages in cash or by bank transfer. We observe stable trends before the reform, followed by a reduction in cash payments during the phase-in period and a more pronounced decline once electronic payments become mandatory. The reduction in cash use is almost entirely mirrored by an increase in bank transfers.¹¹ Panel (b) similarly shows a substantial shift toward lower cash intensity across sectors after the reform.

Because exposure is continuous, the coefficients compare a sector where all workers were paid in cash before the reform with a sector where none were. To illustrate the economic magnitudes, throughout this section we discuss effects for sectors differing by 50 percentage points in pre-reform cash intensity.

Firm behavior We then turn to the effects on firms using administrative payroll records. Panel (a) of Figure 2 shows that firms in more cash-intensive sectors become significantly less likely to remain formally active after the reform. Trends are stable before the reform, but diverge once the phase-in begins. In the long run, firms in sectors with 50 percentage points higher cash intensity are about 7.8 percentage points less likely to be active. Panel A of Table 1 summarizes this and the other firm-level results.

¹⁰ Survey regressions include age-category and education-level indicators and industry-by-quarter and region-by-quarter fixed effects. To improve precision, the survey event studies shown in the appendix use six-month intervals, omitting the period covering 2015Q2–2015Q3.

¹¹ Appendix Figure B.III shows the corresponding raw time series.

Among firms that remain active throughout the sample, we also find a decline in employment and an increase in reported wages. Panels (b) and (c) show that firms in more cash-intensive sectors employ fewer workers but report higher average wages. For the same 50-percentage-point difference in exposure, employment falls by about 5.7 log points and reported wages rise by 4.6 log points in the long run.¹²

To assess whether the increase in reported wages reflects greater labor supply rather than reduced underreporting, we examine hours worked. We find no detectable response in either administrative records or the household survey (Table 1, Panels A and C). Figure 2, panel (d), shows the administrative event study; the corresponding survey event study appears in panel (e) of Appendix Figure B.I. Together with the decline in self-reported wage underreporting, these results support a reporting response rather than an increase in hours worked.

We next ask whether these effects differ by firm size. Informality and collusive wage underreporting are more common among smaller firms (Ulyssea, 2018; Feinmann et al., 2024), which may therefore be particularly affected by restrictions on cash payments. Figure 3 splits firms according to whether they employed fewer than 20 workers before the reform. The decline in formal activity and the increase in reported wages are concentrated among small firms. Employment, however, falls by similar amounts in both groups.

To assess the combined effect of these adjustments, we examine total payroll relative to each firm’s average pre-reform payroll, including zeros when firms become inactive.¹³ Panel (e) of Figure 2 shows that payroll falls in more cash-intensive sectors, despite the rise in reported average wages. The long-run estimate implies a decline equivalent to 13.4 percent of baseline payroll for a 50-percentage-point difference in cash intensity. Thus, the loss of firms and workers from formal payroll records more than offsets the increase in reported wages. We return to the implications for tax revenue in Section 6.

Worker flows and wage composition Using linked worker–firm records, we decompose changes in employment, wages, and payroll to better understand the adjustments underlying our main results. We separate quarterly employment growth into hiring and separation contributions, and changes in average wages relative to 2015Q3 into wage changes among continuing workers and composition effects from departing and entering workers. We also decompose changes in payroll relative to its pre-reform average into payments to continuing workers, baseline payroll lost through departures, and payments to entrants. The employment and wage decompositions use firms active throughout the sample, while the payroll

¹² Results are similar when we include firms that are not active in every quarter; see Appendix Figure B.IX.

¹³ We winsorize relative payroll at the 99.5th percentile.

decomposition also includes firms that become inactive.¹⁴ Panel B of Table 1 reports the results, and Appendix G provides formal definitions.

The results point to lower hiring as the main source of weaker employment growth in more cash-intensive sectors: for a 50-percentage-point difference in exposure, the hiring contribution falls by 1.6 log points, while the separation contribution offsets this by 0.6 log points and is not statistically significant.¹⁵ Most of the increase in reported wages comes from workers not on the firm’s payroll at baseline: the entry component accounts for 2.8 log points of the 4.9-log-point increase in the decomposition sample, compared with 1.2 log points from continuing-worker wages and 1.0 log point from departures. Entry into payroll records, however, may reflect either a new worker or the formalization of an existing employment relationship. These results associate the administrative wage increase mainly with changes in the recorded workforce, while the wage change among continuing workers is imprecisely estimated. The survey evidence below separately assesses whether wage underreporting declined.

Finally, the payroll decomposition attributes the largest share of the decline to departing baseline workers: for a 50-percentage-point difference in cash intensity, departures account for a loss equivalent to 6.4 percent of baseline payroll, changes in payments to continuing workers for another 4.2 percent, and entrants for 2.9 percent, with the latter estimate not statistically significant.

Survey data results Administrative records alone cannot tell us whether firms that disappear have closed or moved into informality, or whether higher reported wages reflect higher earnings or less underreporting. To shed light on these mechanisms, we turn to the household survey. Figure 4 presents the survey results, and Panel C of Table 1 reports the long-run estimates. Given the lower statistical power of the survey data, we bin event periods to improve precision and avoid clutter when displaying overall effects alongside heterogeneity by firm size. Appendix Figures B.I and B.VIII present the more disaggregated event studies at six-month intervals. Panel (a) of Figure 4 shows that the reduction in the share of workers paid in cash is more pronounced in small firms.

Among formally employed workers, panel (b) shows no significant increase in survey-reported earnings after the reform. This contrasts with the increase in administrative wages and is consistent with improved reporting to the tax authority. We also look at more direct evidence coming from workers’ responses about wage underreporting. Panel (d) of Figure 4

¹⁴ The wage decomposition additionally requires at least one worker shared between the baseline and each comparison quarter.

¹⁵ This exercise describes hiring and separations late in the sample rather than their contribution to the cumulative employment decline.

shows that formal workers in more cash-intensive sectors become less likely to report that their pension contributions are based on only part of their earnings. For a 50-percentage-point difference in cash intensity, the long-run estimate in Panel C of Table 1 implies that underreporting falls by about 1.8 percentage points. Although this question does not reveal the amount underreported, the result supports the interpretation that reduced evasion contributes to the increase in wages reported in administrative records.

At the same time, panel (c) shows that informality rises. The corresponding long-run estimate in Table 1 implies that workers in sectors with 50 percentage points higher cash intensity become about 5.1 percentage points more likely to be informally employed. Panel (f) of Figure 4 shows no significant change in employment among respondents whose current or previous sector can be identified.¹⁶ Together, these findings suggest that part of the decline in registered employment reflects a shift toward informality. They do not establish that overall employment is unaffected, since the employment analysis excludes respondents without an identifiable sector.

The survey results by firm size are also consistent with this interpretation. Underreporting falls primarily among workers in small establishments, where we observe the increase in administrative wages. Survey-reported wages show no significant change in either size group, while informality rises in both.¹⁷ Taken together, the results suggest that the reform reduced underreporting among formal workers while encouraging some activity to move into full informality, although we cannot directly follow these transitions in the survey.

Alternative reform channels An alternative explanation is that the results reflect other components of the financial inclusion reform, such as VAT rebates for debit-card purchases or greater access to banking services. For instance, an increase in debit-card purchases could reduce the cash available to businesses, affecting their wage payments and formal activity independently of the wage-payment mandate.

Several findings weigh against this explanation. First, debit-card use rose immediately after the VAT rebates were introduced in August 2014 (Brockmeyer and Sáenz Somarriba, 2025), more than a year before the wage-payment phase-in began. If this were driving our results, we would expect a break around that date, but the event studies show no such pattern. Second, increased debit-card purchases should matter most for sectors such as retail and restaurants, whereas our design uses variation across a much broader set of activities. Results remain qualitatively similar when we exclude these sectors, as shown in Appendix C. Finally, Brockmeyer and Sáenz Somarriba (2025) find little evidence that the increase in

¹⁶ Unemployed respondents cannot be assigned a current firm size, so we cannot estimate employment effects separately by firm size.

¹⁷ Appendix Figure B.VIII reports the corresponding event studies.

debit-card use changed VAT reporting, suggesting a limited role for this channel in explaining our findings.

5.2 Robustness checks

We assess whether the results depend on particular industries, the definition of exposure, or the sample of firms. Appendix Figures D.I and D.II compare the long-run estimates after excluding each broad industry in turn. We also replace continuous cash intensity with an indicator for sectors above the median within their industry, so that the comparison is between relatively high- and low-cash-intensity sectors (Appendix Figures E.I and E.II). Finally, Appendix Figure B.IX shows that the employment and wage results are similar when we relax the requirement that firms remain active throughout the sample.

6. Revenue implications

Our results suggest two opposing effects on payroll-tax revenue. On the one hand, reduced wage underreporting increases the reported tax base. On the other hand, the disappearance of firms and workers from formal payroll records reduces it. We use a simple back-of-the-envelope calculation to assess which effect dominates.

Assuming a constant proportional contribution rate τ , revenue is $T = \tau P$, where P denotes total reported payroll. A given percentage change in payroll therefore translates into the same percentage change in revenue. Our payroll measure combines the effects on firm activity, employment, and reported wages, including the loss of payroll when a firm becomes inactive.

The long-run payroll coefficient is -0.2687 . In 2014, 37 percent of formal workers were paid in cash, so using this share to approximate economy-wide exposure gives $-0.2687 \times 0.37 \approx -0.0994$, or a revenue loss equivalent to 9.9 percent of the pre-reform revenue base. This calculation assumes that the reform has no effect in sectors with no initial cash use and that the average firm response provides a reasonable approximation to the aggregate payroll response. Because the latter assumption may be restrictive, given substantial differences across firms in their contribution to total payroll, we also estimate a specification that weights firms by their average pre-reform payroll. The resulting coefficient is -0.205 , while average cash intensity weighted by baseline payroll is 0.365, implying a loss of 7.5 percent of baseline revenue.

These calculations suggest that the loss of formal firms and jobs may more than offset the revenue gains from higher reported wages. The estimates should not be interpreted as a

precise measure of the change in aggregate tax revenue. The weighted estimate is less precise, the calculations cover a relatively short post-reform horizon, and they extrapolate from firms present before the reform without incorporating the contribution of new firms. Rather, the exercise illustrates an important potential fiscal consequence of policies that restrict cash payments in an effort to improve compliance. Even when such policies improve reporting among firms that remain formal, the resulting revenue gains may be offset if some firms and workers respond by moving out of the formal sector.

7. Conclusion

This paper studies Uruguay’s mandate requiring wages to be paid electronically rather than in cash. Exploiting cross-sector variation in pre-reform cash use, we combine administrative payroll records and household survey data to estimate the reform’s effects on formal activity, employment, reported wages, underreporting, and informality.

The reform reduced formal firm activity and payroll employment in sectors that had relied more heavily on cash, while reported wages increased among always-active firms. The activity and reported-wage effects were concentrated among smaller firms; employment declines were similar across firm-size groups. Survey evidence helps interpret these patterns: survey-reported wages showed no statistically significant increase, survey-reported wage evasion declined, informality increased, and the overall employment estimate was not statistically distinguishable from zero. Taken together, the results are consistent with reduced partial evasion within formal employment alongside a shift of some activity into full informality.

These findings highlight an important tradeoff in policies that restrict the use of cash to improve compliance. Restricting cash can improve compliance among firms that remain in the formal sector, consistent with evidence that restricting cash availability can reduce tax evasion (Das et al., 2023). Yet these gains may be offset if marginal firms respond by leaving the formal sector altogether. The decline in payroll relative to its pre-reform mean illustrates this tradeoff. Our back-of-the-envelope calculations suggest a loss in payroll-tax revenue of roughly 7 to 10 percent of the pre-reform revenue base, depending on whether firms are weighted by baseline payroll. These estimates are illustrative rather than precise measures of the aggregate revenue effect, but they show that losses of formal activity and jobs can be large enough to outweigh gains from improved wage reporting.

The reform also expanded the use of bank accounts for wage payments and may have generated broader benefits from financial inclusion that are outside the scope of this paper. Our results therefore do not provide a complete welfare evaluation. They instead show that policies promoting electronic payments can change not only the level of noncompliance, but

also the form it takes.

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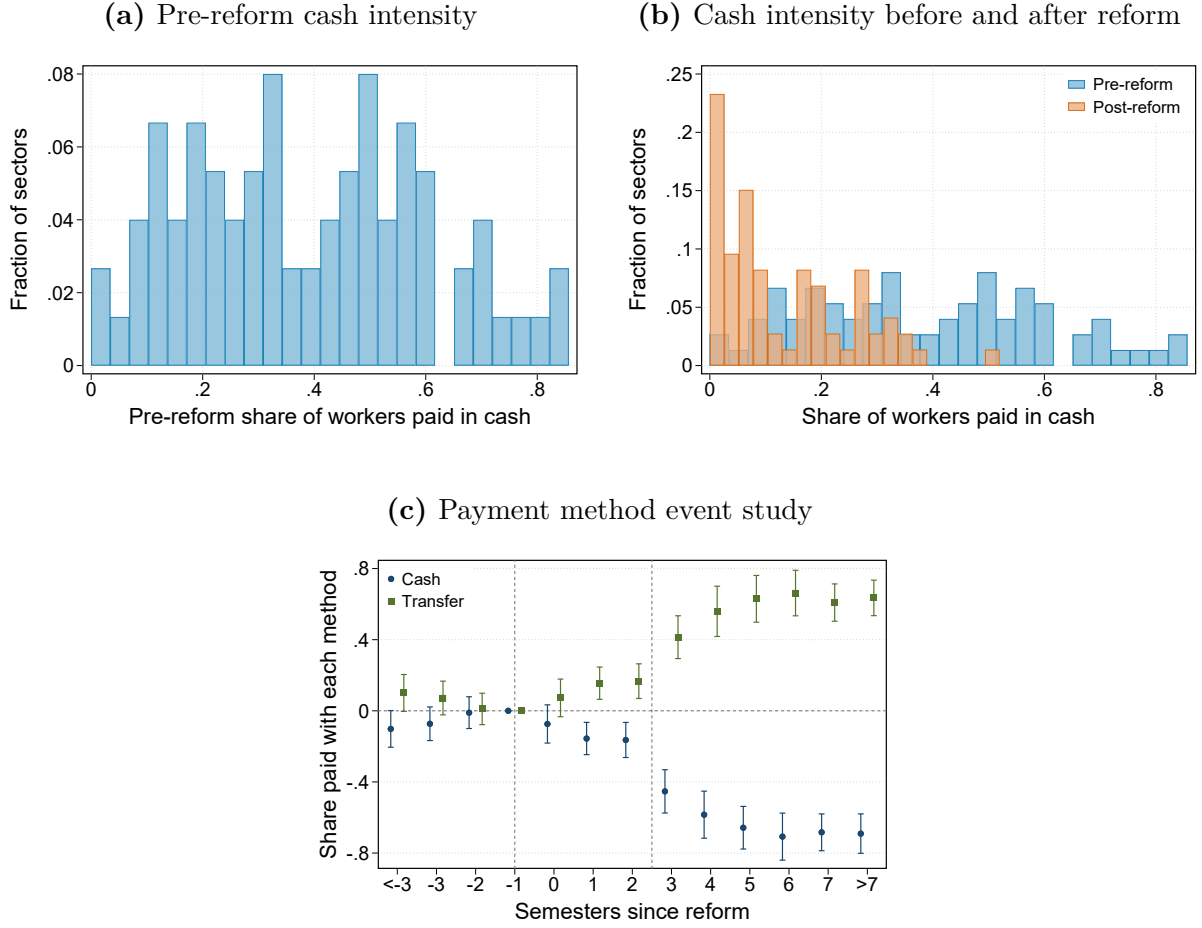
Tables and Figures

Table 1: Long-run effects of the wage-payment reform

Panel A. Administrative data: firm-level outcomes										
	Overall					By firm size				
	Active	Employment (log)	Average wage (log)	Hours	Relative payroll	Active	Employment (log)	Average wage (log)	Hours	Relative payroll
Cash Int. $\times$ Long run	-.157*** (.052)	-.113* (.0611)	.0915** (.0401)	3.020 (3.174)	-.269*** (.0736)					
Small firm $\times$ Cash Int. $\times$ Long run						-.161*** (.0515)	-.113* (.0614)	.096** (.0395)	2.884 (3.146)	-.271*** (.0728)
Large firm $\times$ Cash Int. $\times$ Long run						.0454 (.055)	-.122* (.0644)	.0169 (.044)	5.304 (3.885)	-.164** (.0778)
Observations	814,968	560,832	559,373	559,981	814,968	814,968	560,832	559,373	559,981	814,968
Sectors	73	73	73	73	73	73	73	73	73	73
Panel B. Employment, wage, and payroll accounting										
	Quarterly log-employment growth		Change in log average wages			Relative payroll				
	Hiring component	Separation component	Stayer wage component	Departure component		Entry component	Stayer wage component	Departure component	Entry component	
Cash Int. $\times$ Long run	-0.031** (0.015)	0.013 (0.013)	0.024 (0.042)	0.020 (0.018)	0.056*** (0.018)	-0.083*** (0.027)	-0.128** (0.057)	-0.057 (0.101)		
Observations	537,464	537,464	515,376	515,376	515,376	814,968	814,968	814,968		
Sectors	73	73	73	73	73	73	73	73		
Panel C. Survey data										
	Overall						By firm size			
	Earnings (log)	Informal	Under-reports	Paid in cash	Employed	Hours	Earnings (log)	Informal	Under-reports	Hours
Cash Int. $\times$ Long run	0.012 (0.049)	0.103*** (0.027)	-0.036** (0.015)	-0.660*** (0.050)	-0.001 (0.023)	-0.494 (0.834)				
Small firm $\times$ Cash Int. $\times$ Long run							0.024 (0.049)	0.082*** (0.022)	-0.051*** (0.014)	-0.516 (0.839)
Large firm $\times$ Cash Int. $\times$ Long run							-0.007 (0.053)	0.070** (0.028)	0.001 (0.016)	-0.320 (0.808)
Observations	135,336	248,076	139,598	135,310	406,097	177,676	135,336	248,076	139,598	177,676
Sectors	73	73	73	73	73	73	73	73	73	73

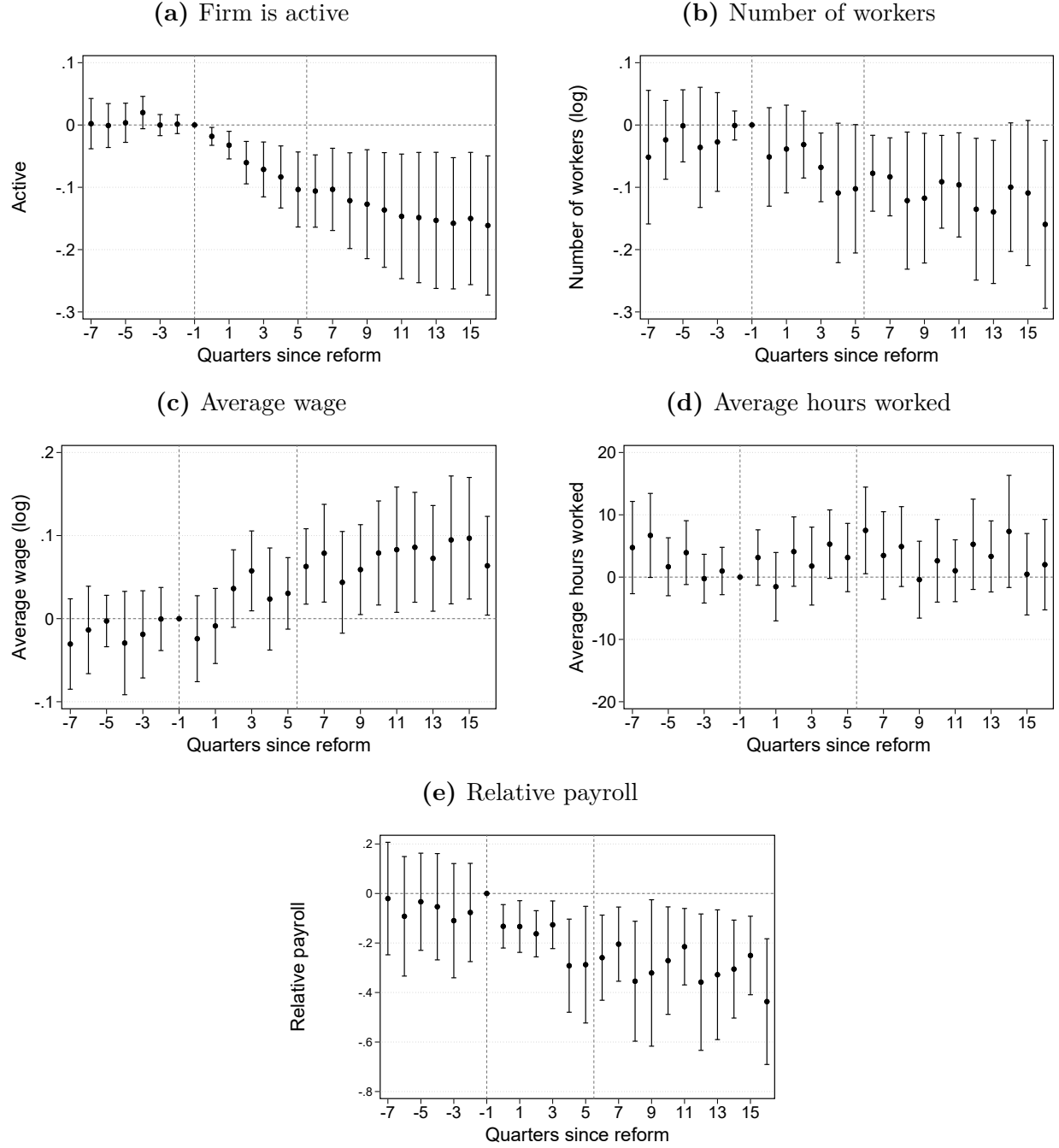
Notes: This table reports long-run difference-in-differences coefficients from the pooled version of equation (7). Cash Int. is pre-reform cash intensity at the sector level. Long run is an indicator equal to one in the final three quarters of the sample (2019Q2–2019Q4). Panels A and B use administrative data; Panel C uses household survey data. In Panel A, the dependent variables are an indicator for whether the firm is active, log employment, log average reported wages, average monthly hours worked, and payroll relative to its pre-reform average, respectively; the same outcomes are repeated in the firm-size columns. In Panel B, the first two columns report the hiring and signed separation components of quarterly log-employment growth; the next three report the stayer wage, departure, and entry components of the change in log average wages; and the final three report the corresponding components of relative payroll. In Panel C, the dependent variables are log earnings, indicators for informal employment, wage underreporting, cash wage payments, employment, and weekly hours worked, respectively; the firm-size columns repeat earnings, informal employment, wage underreporting, and weekly hours worked. Small firms are defined as firms with fewer than 20 employees. Standard errors clustered at the sector level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure 1: Sector-level cash intensity and payment-method response



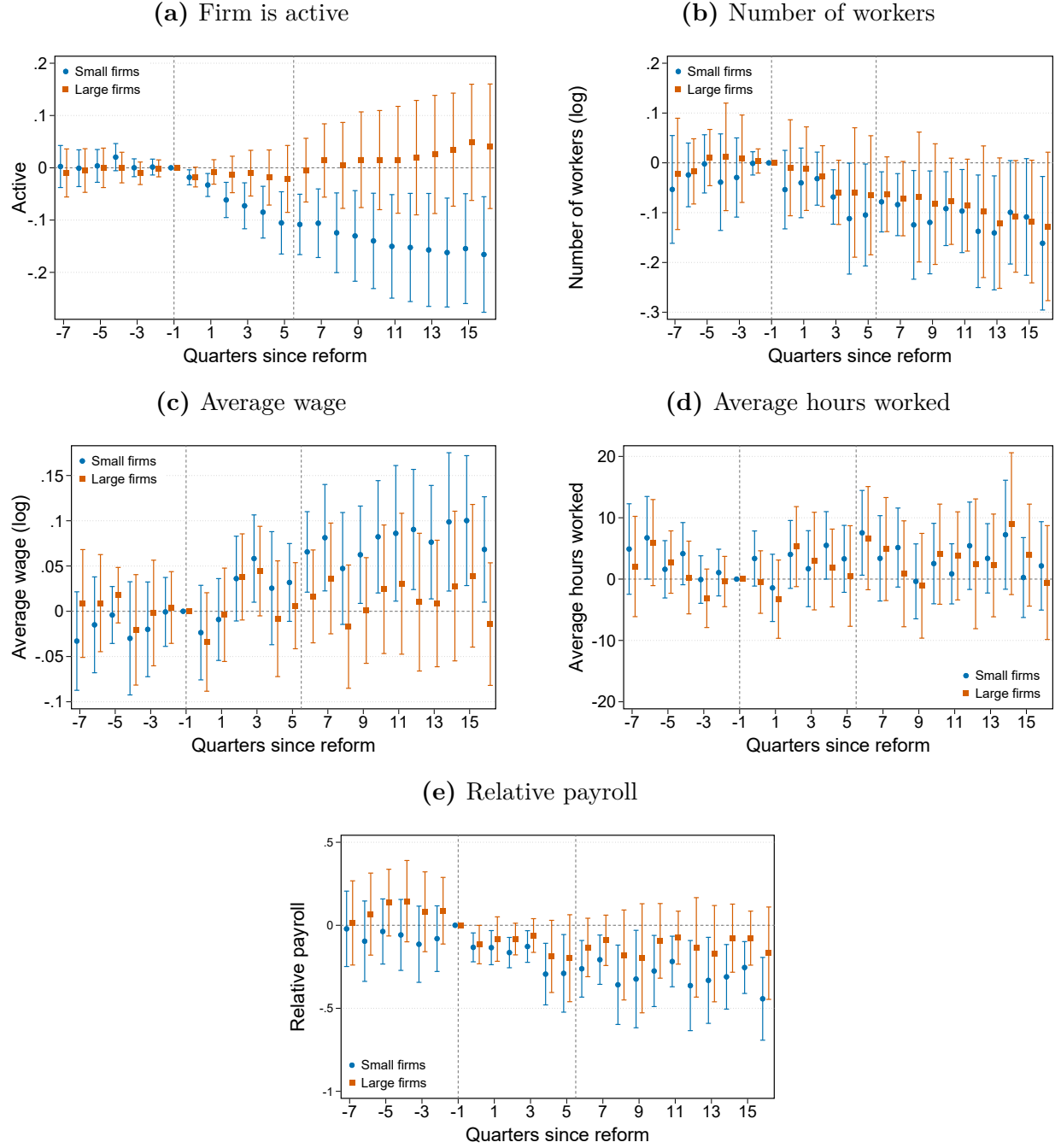
Notes: Panels (a) and (b) report histograms of sector-level cash intensity. Panel (a) shows the pre-reform period and panel (b) shows the pre-reform period in blue and the post-reform period in orange. Panel (c) reports event-study estimates for cash payments (blue) and bank transfers (green). Dashed lines mark the phase-in and mandate periods. Bars show 95% confidence intervals using sector-clustered standard errors.

Figure 2: Main effects of the reform (administrative data)



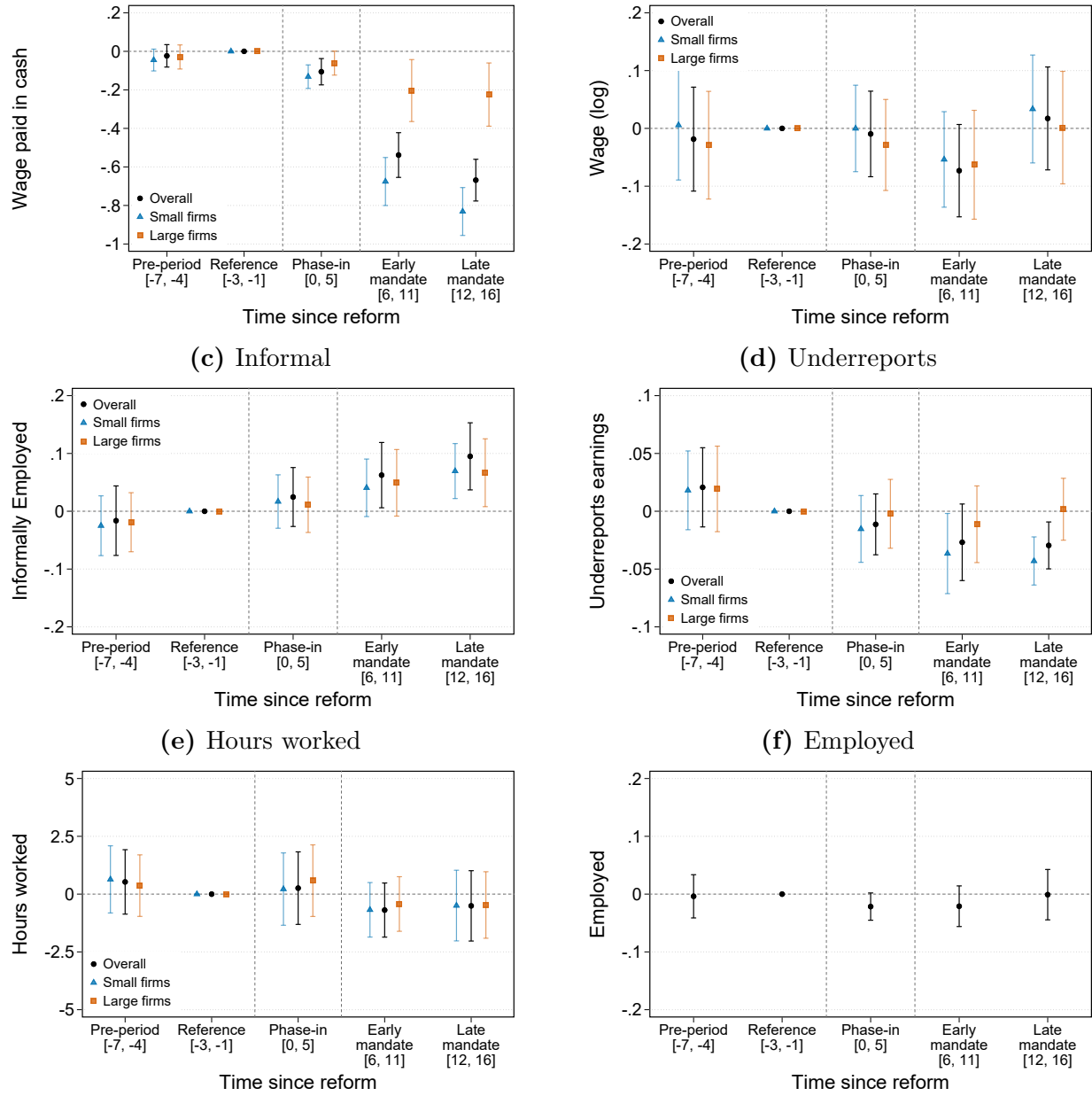
Notes: This figure reports event-study estimates from equation 7. Dashed lines mark the phase-in and mandate periods. In panel (a), the dependent variable is an indicator equal to 1 if the firm is active in administrative records and zero otherwise. In panel (b), the dependent variable is the log of the number of workers employed at the firm in administrative records. In panel (c), the dependent variable is the log of the average wage reported for employees at the firm in administrative records. In panel (d), the dependent variable is the average monthly hours worked by employees at the firm in administrative records. In panel (e), the dependent variable is payroll relative to average pre-reform payroll. Vertical bars show 95% confidence intervals using sector-clustered standard errors.

Figure 3: Main effects of the reform - heterogeneity by firm size (administrative data)



Notes: This figure reports event-study estimates from equation 7 by firm size. Small firms have fewer than 20 pre-reform employees. Dashed lines mark the phase-in and mandate periods. In panel (a), the dependent variable is an indicator equal to 1 if the firm is active in administrative records and zero otherwise. In panel (b), the dependent variable is the log of the number of workers employed at the firm in administrative records. In panel (c), the dependent variable is the log of the average wage reported for employees at the firm in administrative records. In panel (d), the dependent variable is the average monthly hours worked by employees at the firm in administrative records. In panel (e), the dependent variable is payroll relative to average pre-reform payroll. Vertical bars show 95% confidence intervals using sector-clustered standard errors.

Figure 4: Main effects of the reform and heterogeneity by firm size (survey data)
 (a) Paid in cash (b) Wage



Notes: This figure reports survey-weighted estimates from the pooled version of equation 7. In panel (a), the outcome is a dummy variable equal to 1 if the wage is paid in cash and zero otherwise. In panel (b), the outcome is the log of earnings. In panel (c), the outcome is a dummy variable equal to 1 if the worker is informally employed and zero otherwise. In panel (d), the outcome is a dummy variable equal to 1 if the worker underreports their labor earnings for tax purposes and zero otherwise. In panel (e), the outcome is weekly hours worked. In panel (f), the outcome is a dummy variable equal to 1 if the worker is employed and zero otherwise. Black circles show overall estimates, blue triangles small-firm estimates (fewer than 20 employees), and orange squares large-firm estimates. Coefficients interact pre-reform sector cash intensity with the quarter bins shown on the horizontal axis; quarters -3 to -1 are omitted. Dashed lines separate the pre-reform, phase-in, and mandate periods. Capped vertical bars represent 95% confidence intervals from standard errors clustered at the sector level.

Cash Wage Payments, Tax Evasion, and Informality

Supplemental Appendix

Marcelo Bérigolo¹, Javier Feinmann², Maximiliano Lauletta³

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A. Tables

Table A.I: Summary statistics

	Observations	Mean	Standard deviation	Median
<i>Panel A. Administrative data</i>				
Active	814,968	0.853	0.354	1.000
Number of workers	560,832	19.450	106.357	7.000
Average wage	559,373	31,300.429	24,569.121	24,672.521
Monthly hours worked	559,981	196.383	65.718	213.200
Small firm	814,968	0.871	0.335	1.000
<i>Panel B. Household survey data</i>				
Paid in cash	135,310	0.259	0.438	0.000
Female	248,076	0.395	0.489	0.000
Age	248,076	40.684	13.623	40.000
Informally employed	248,076	0.277	0.447	0.000
Underreports earnings	139,598	0.040	0.195	0.000
Wage	135,336	26,951.516	16,757.729	22,000.000
Weekly hours worked	177,676	42.498	12.097	44.000
Small firm	248,076	0.641	0.480	1.000

Notes: This table reports summary statistics for the corresponding outcome-specific estimation samples; firm size uses the activity sample, and survey demographics and firm size use the informality sample. Survey moments use sampling weights; observation counts are unweighted. Panel A reports summary statistics for administrative data. Active is an indicator equal to 1 if the firm is active in a given period and zero otherwise. Number of workers is the total number of dependent workers employed by a firm in a given period. Average wage is the firm's average wage in September 2015 pesos. Monthly hours worked is the average per employee at the firm. Small firm is an indicator equal to 1 if the firm had fewer than 20 employees on average during the pre-reform period. Panel B reports summary statistics for household survey data. Paid in cash is an indicator equal to 1 if the worker reports being paid in cash. Female is a dummy variable equal to 1 if the respondent is female. Age is the age of the respondent in years. Informally employed is a dummy variable equal to 1 if the worker is informally employed (does not make pension contributions). Underreports earnings is a dummy variable equal to 1 if the respondent states that they underreport their labor earnings for tax purposes. Wage is the workers' earnings in current Uruguayan pesos. Weekly hours worked refers to the respondent's main occupation. Small firm is an indicator equal to 1 if the firm has fewer than 20 employees.

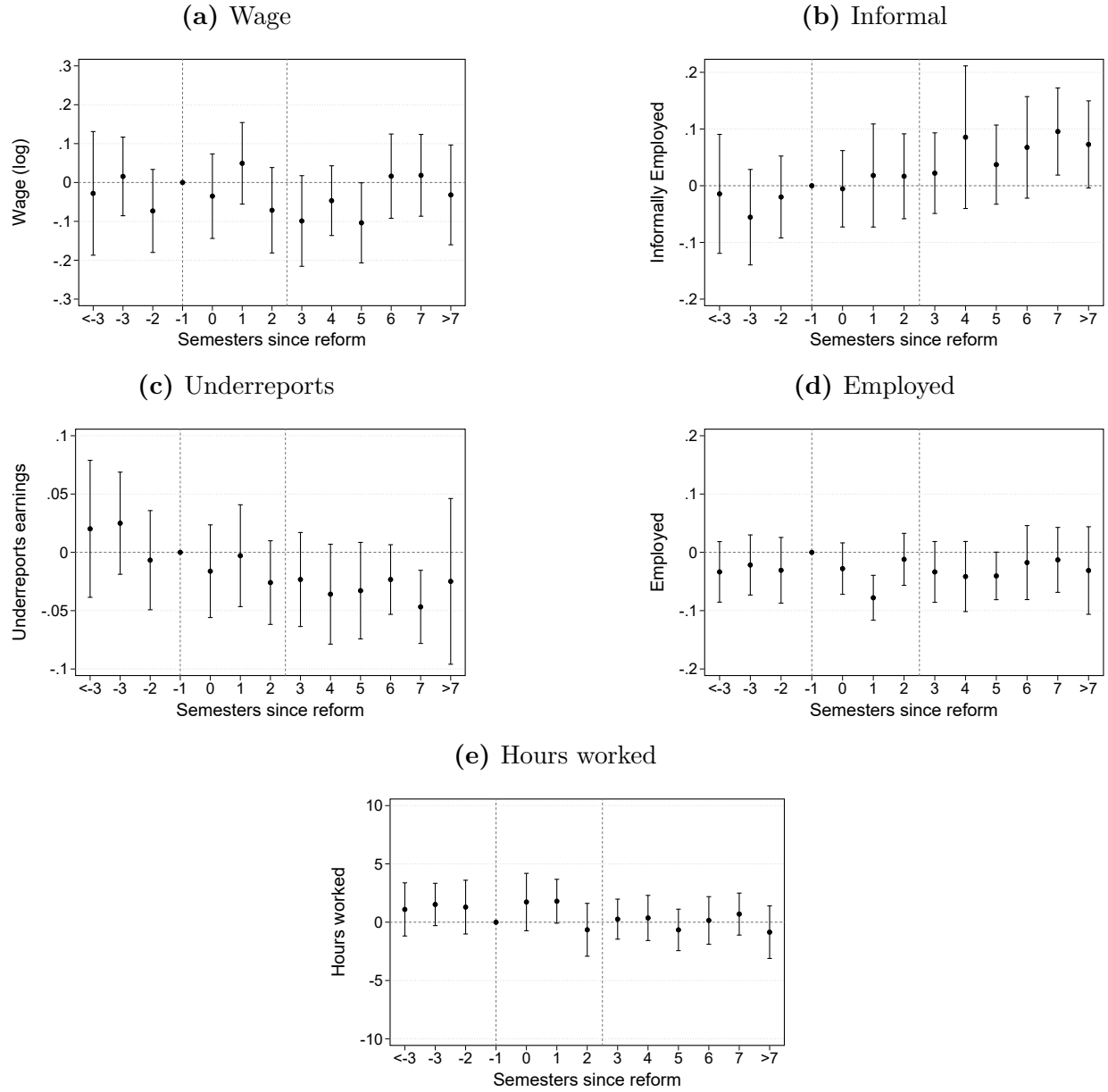
Table A.II: Long-run effects in household survey data: unweighted regressions

	Overall						By firm size			
	Earnings (log)	Informal	Under- reports	Paid in cash	Employed	Hours	Earnings (log)	Informal	Under- reports	Hours
Cash Int. $\times$ Long run	0.015 (0.051)	0.095*** (0.032)	-0.039** (0.017)	-0.674*** (0.051)	0.002 (0.025)	-0.725 (0.810)				
Small firm $\times$ Cash Int. $\times$ Long run							0.024 (0.050)	0.077*** (0.025)	-0.054*** (0.016)	-0.769 (0.811)
Large firm $\times$ Cash Int. $\times$ Long run							0.000 (0.055)	0.075** (0.031)	-0.001 (0.018)	-0.515 (0.780)
Observations	135,336	248,076	139,598	135,310	406,097	177,676	135,336	248,076	139,598	177,676
Sectors	73	73	73	73	73	73	73	73	73	73

Notes: This table reports long-run difference-in-differences coefficients from the pooled version of equation (7) using household survey data without sampling weights. Cash Int. is pre-reform cash intensity at the sector level. Long run is an indicator equal to one in the final three quarters of the sample (2019Q2–2019Q4). The dependent variables in the first six columns are log earnings, indicators for informal employment, wage underreporting, cash wage payments, and employment, and weekly hours worked, respectively; the firm-size columns repeat earnings, informal employment, wage underreporting, and weekly hours worked. Small firms are defined as firms with fewer than 20 employees. Specifications and samples are otherwise identical to Panel C of Table 1. Standard errors clustered at the sector level are reported in parentheses. $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

B. Additional figures

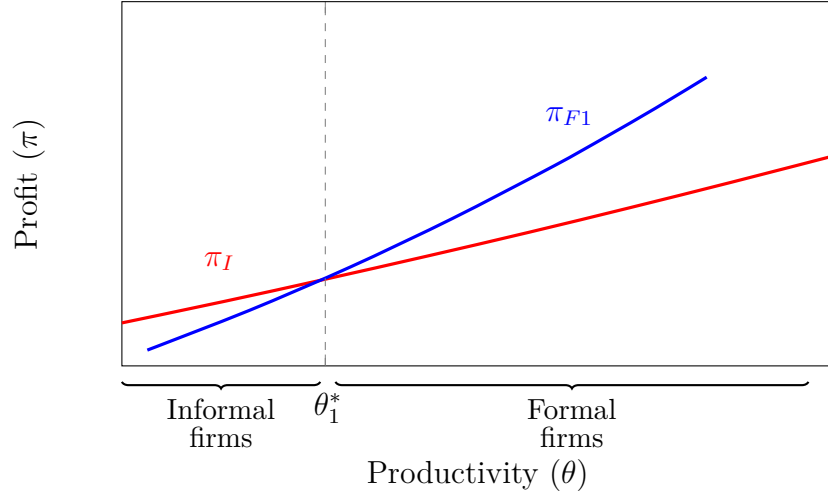
Figure B.I: Main effects of the reform (survey data)



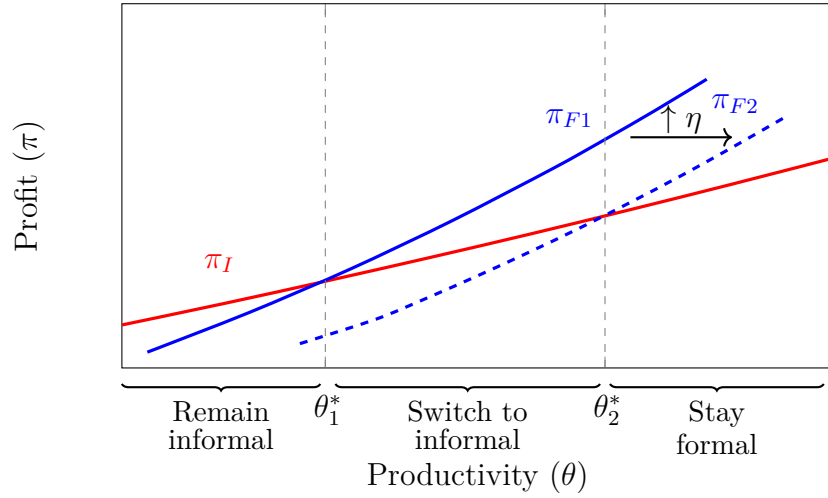
Notes: This figure shows OLS estimates of the β_k coefficients from equation 7 grouped into six-month periods. In panel (a) the dependent variable is the natural logarithm of the wage. In panel (b) the dependent variable is an indicator equal to 1 if the worker is informally employed. In panel (c) the dependent variable is an indicator equal to 1 if the worker underreports their labor earnings for tax purposes. In panel (d) the dependent variable is an indicator equal to 1 if the worker is employed and 0 otherwise. In panel (e) the dependent variable is weekly hours worked among formally employed private-sector workers aged 18 or older. The first dashed line indicates the beginning of the phase-in period and the second dashed line indicates the beginning of the mandate period. Capped vertical bars represent 95% confidence intervals from standard errors clustered at the sector level.

Figure B.II: Model figures

(a) Illustration of equilibrium

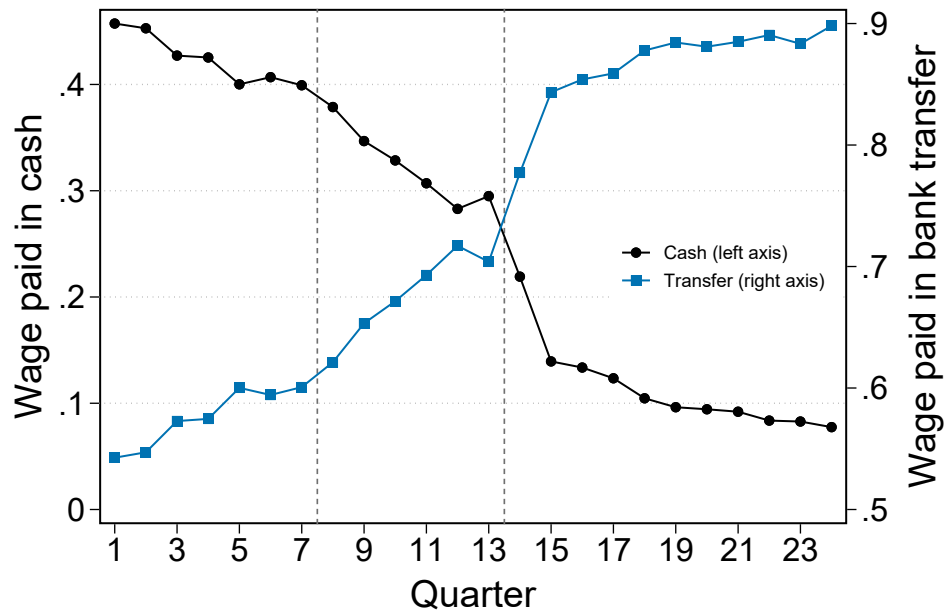


(b) Illustration of a cash ban



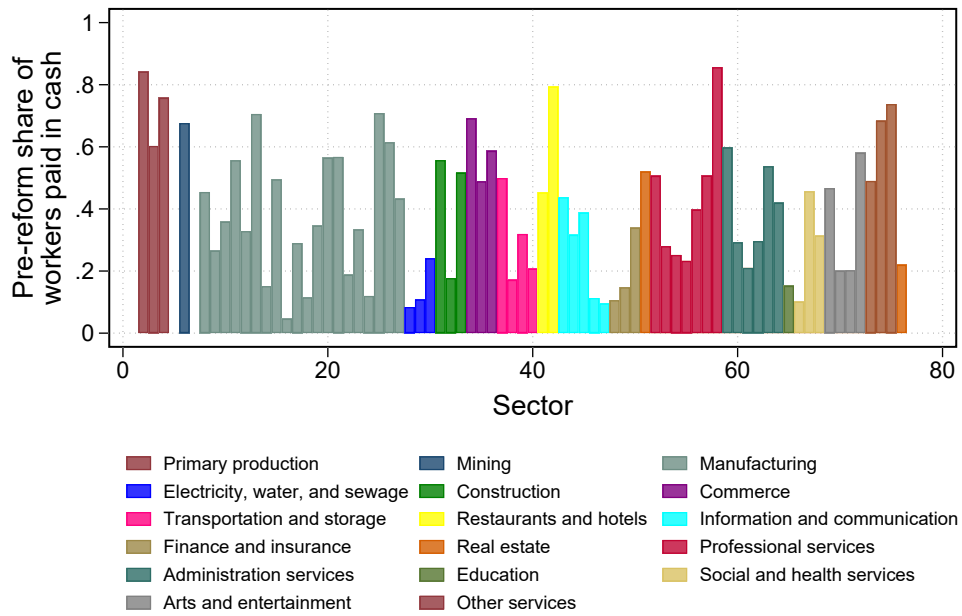
Notes: This figure shows a graphical representation of the equilibrium from our conceptual framework. Panel (a) illustrates the static equilibrium and Panel (b) illustrates the cash wage ban. The horizontal axis represents productivity and the vertical axis represents profits. Blue corresponds to profits for formal firms and red corresponds to profits for informal firms.

Figure B.III: Share of workers paid in cash or by bank transfer over time



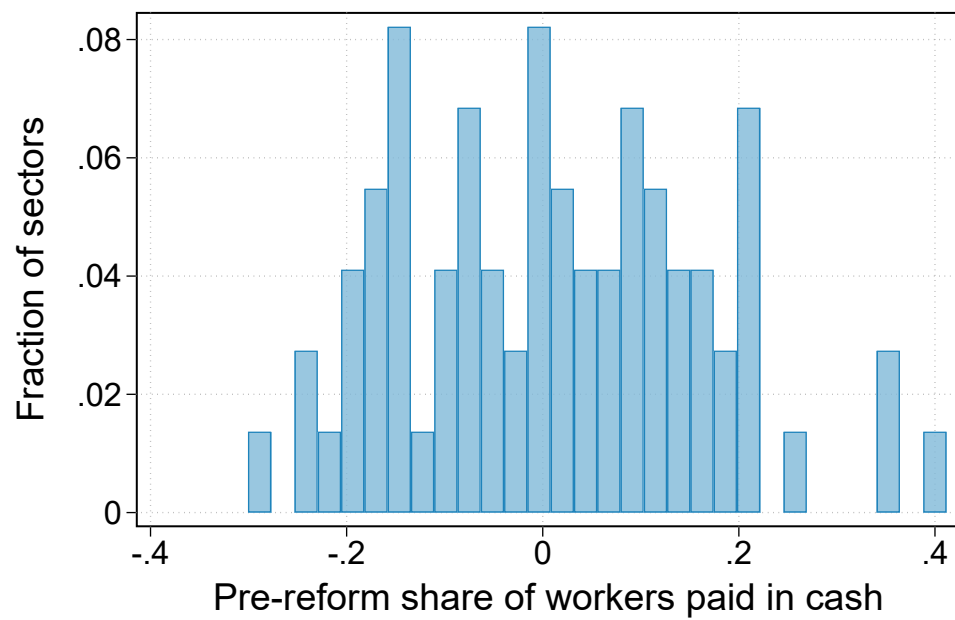
Notes: This figure shows the proportion of formal workers that are paid in cash (in black corresponding to the left axis) and the proportion that are paid with a bank transfer (in blue corresponding to the right axis) in each quarter. The first dashed line indicates the beginning of the phase-in period and the second dashed line indicates the beginning of the mandate period.

Figure B.IV: Within-industry pre-reform cash intensity



Notes: This figure shows the share of workers paid in cash during the pre-reform period for each sector. The bar colors indicate the industry grouping. Sectors are defined at the 2-digit ISIC level and broad industries follow ISIC section groupings.

Figure B.V: Pre-reform cash intensity (residualized on industry fixed effects)



Notes: This figure shows a histogram of the pre-reform sector-level share of workers who are paid in cash, residualized from industry fixed effects. Sectors are defined at the 2-digit ISIC level and broad industries follow ISIC section groupings.

Figure B.VI: Correlations between cash usage, informality, and underreporting

(a) Informal employment

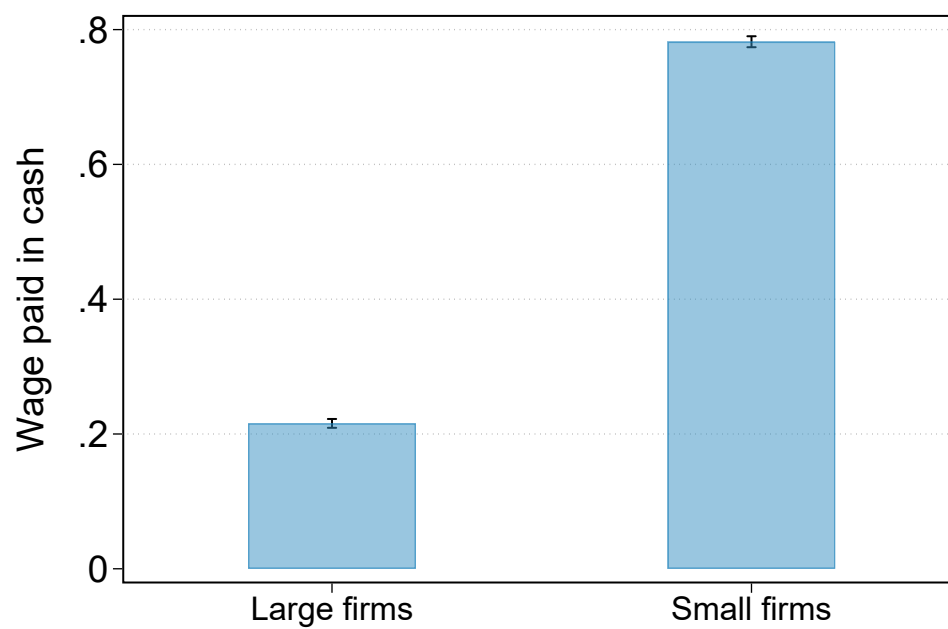


(b) Wage underreporting



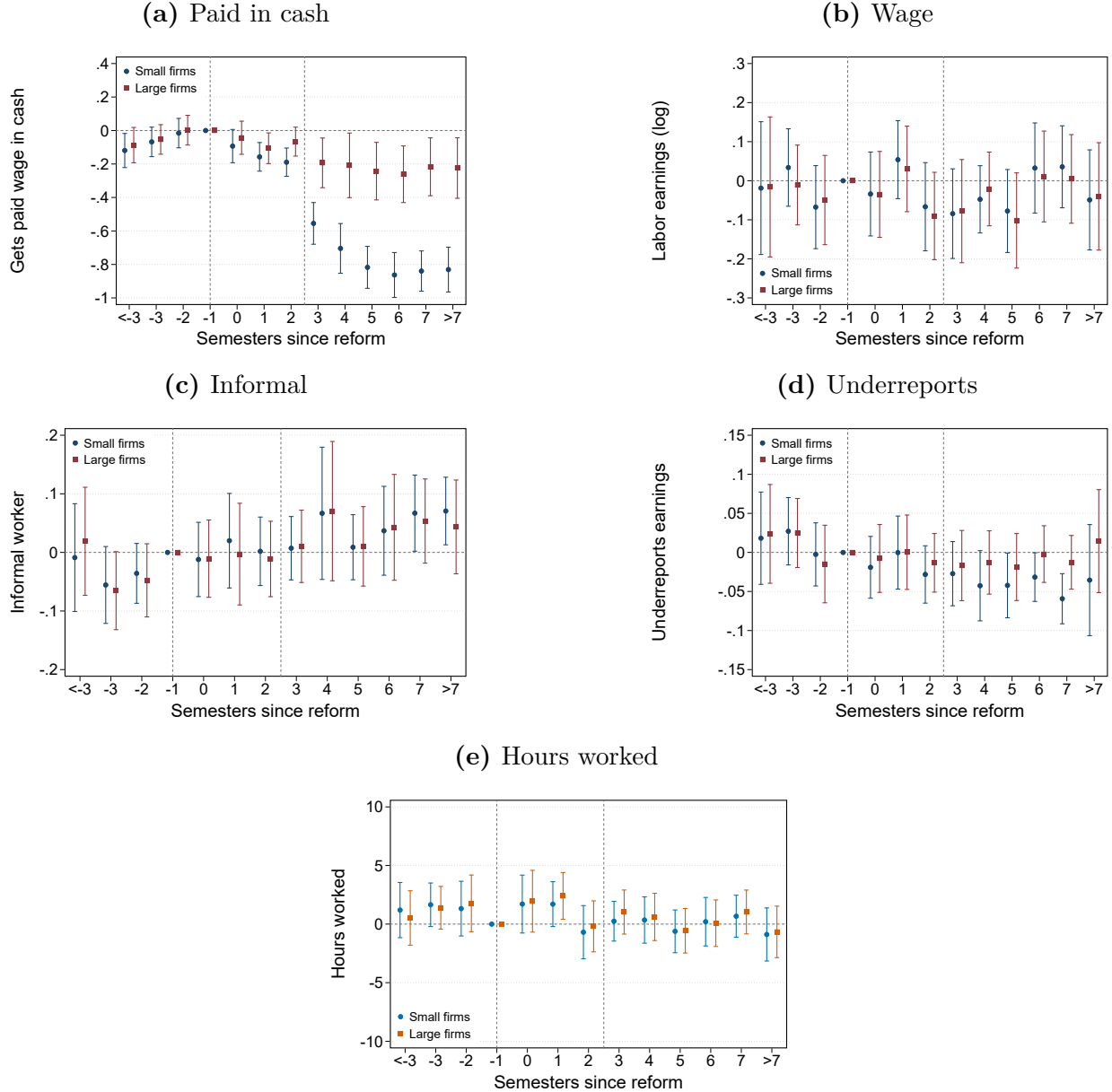
Notes: This figure shows binned scatterplots of the sector-level spread of illegal labor market practices and the share of workers reporting cash payments in 2014. Panel (a) shows the share of workers who are informally employed and panel (b) shows the share of formal employees who admit to underreporting their labor earnings for tax purposes.

Figure B.VII: Proportion of workers paid in cash by firm size



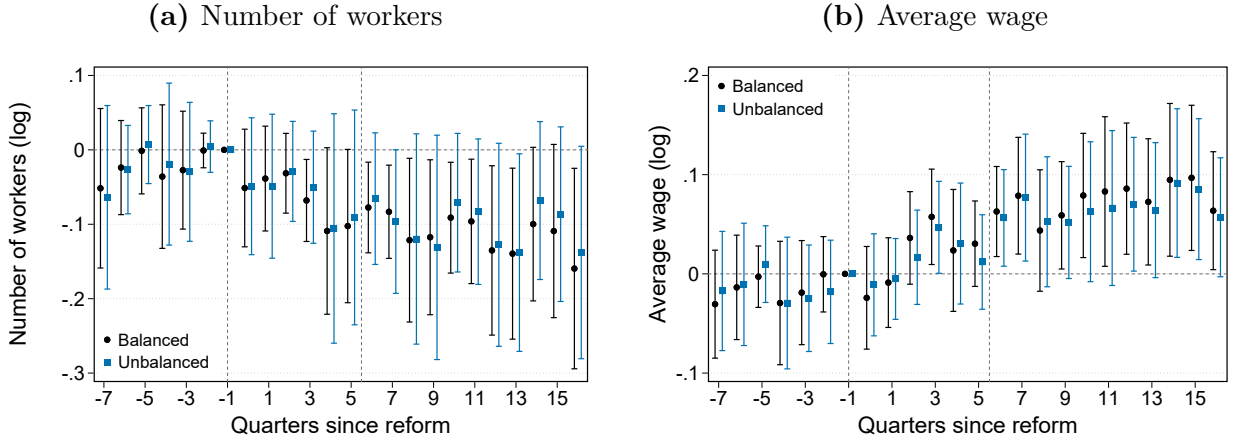
Notes: This figure shows the proportion of formal workers reporting cash payments in 2014, by firm size. Small firm corresponds to firms with fewer than 20 workers and large firm corresponds to 20 or more. Vertical bars represent 95% confidence intervals from robust standard errors.

Figure B.VIII: Heterogeneous effects of the reform - firm size (survey data)



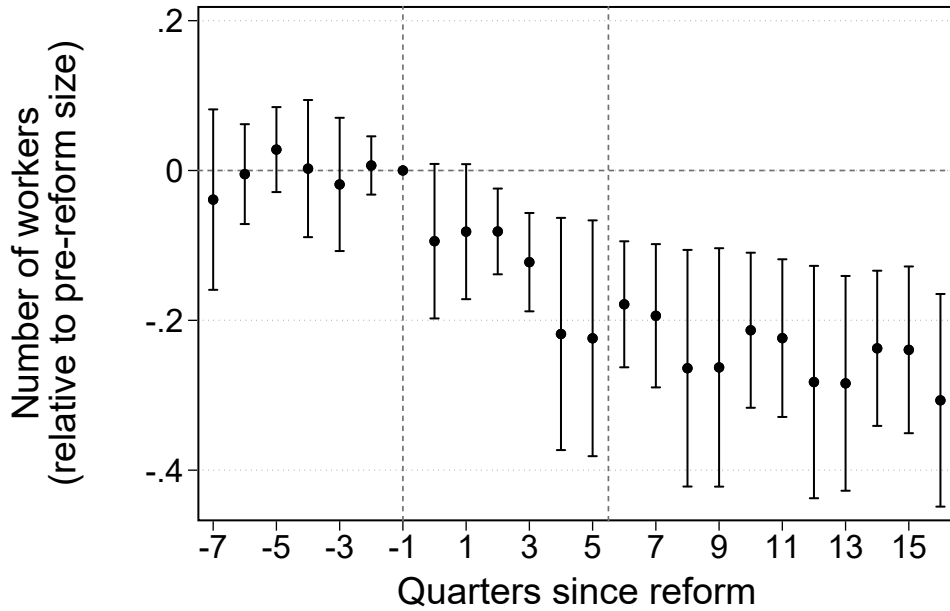
Notes: This figure shows OLS estimates of the β_k coefficients from equation 7 grouped into six-month periods interacted with an indicator for small firm and an indicator for large firm. Blue coefficients correspond to small firms (fewer than 20 employees) and red coefficients (orange in panel (e)) correspond to large firms (20 or more employees). In panel (a) the dependent variable is an indicator equal to 1 if the worker is paid in cash. In panel (b) the dependent variable is the natural logarithm of the wage. In panel (c) the dependent variable is an indicator equal to 1 if the worker is informally employed. In panel (d) the dependent variable is an indicator equal to 1 if the worker underreports their labor earnings for tax purposes. In panel (e) the dependent variable is weekly hours worked among formally employed private-sector workers aged 18 or older. The first dashed line indicates the beginning of the phase-in period and the second dashed line indicates the beginning of the mandate period. Capped vertical bars represent 95% confidence intervals from standard errors clustered at the sector level.

Figure B.IX: Always-active firms and full-sample comparison



Notes: Baseline estimates for always-active firms (black) are compared with estimates using the unbalanced sample (blue). Dashed lines mark the phase-in and mandate periods. Bars show 95% confidence intervals using sector-clustered standard errors.

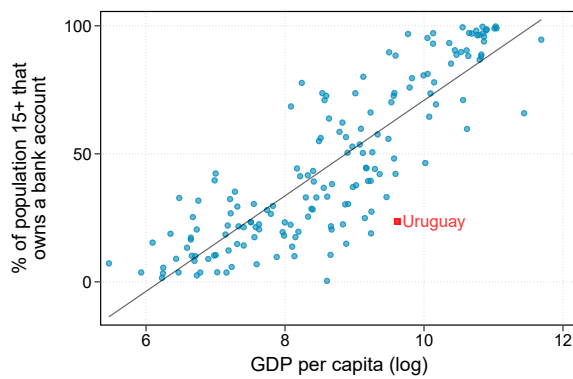
Figure B.X: Number of workers (extensive and intensive margin combined)



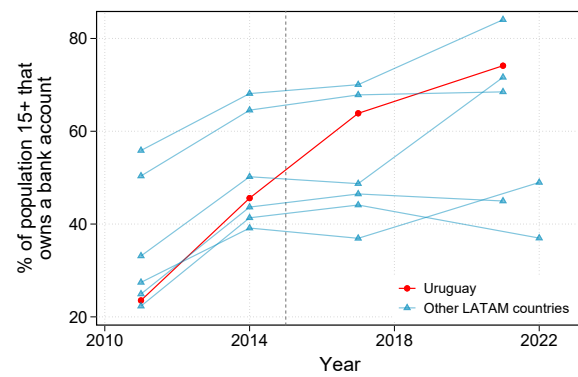
Notes: This figure shows OLS estimates of the β_k coefficients from equation 7. The dependent variable is the number of workers employed in the firm relative to the average number of workers employed during the pre-reform period. Inactive firms are coded as having zero workers. The first dashed line indicates the beginning of the phase-in period and the second dashed line indicates the beginning of the mandate period. Capped vertical bars represent 95% confidence intervals from standard errors clustered at the sector level.

Figure B.XI: Financial inclusion in Uruguay and other countries

(a) Cross-country account ownership (2011)



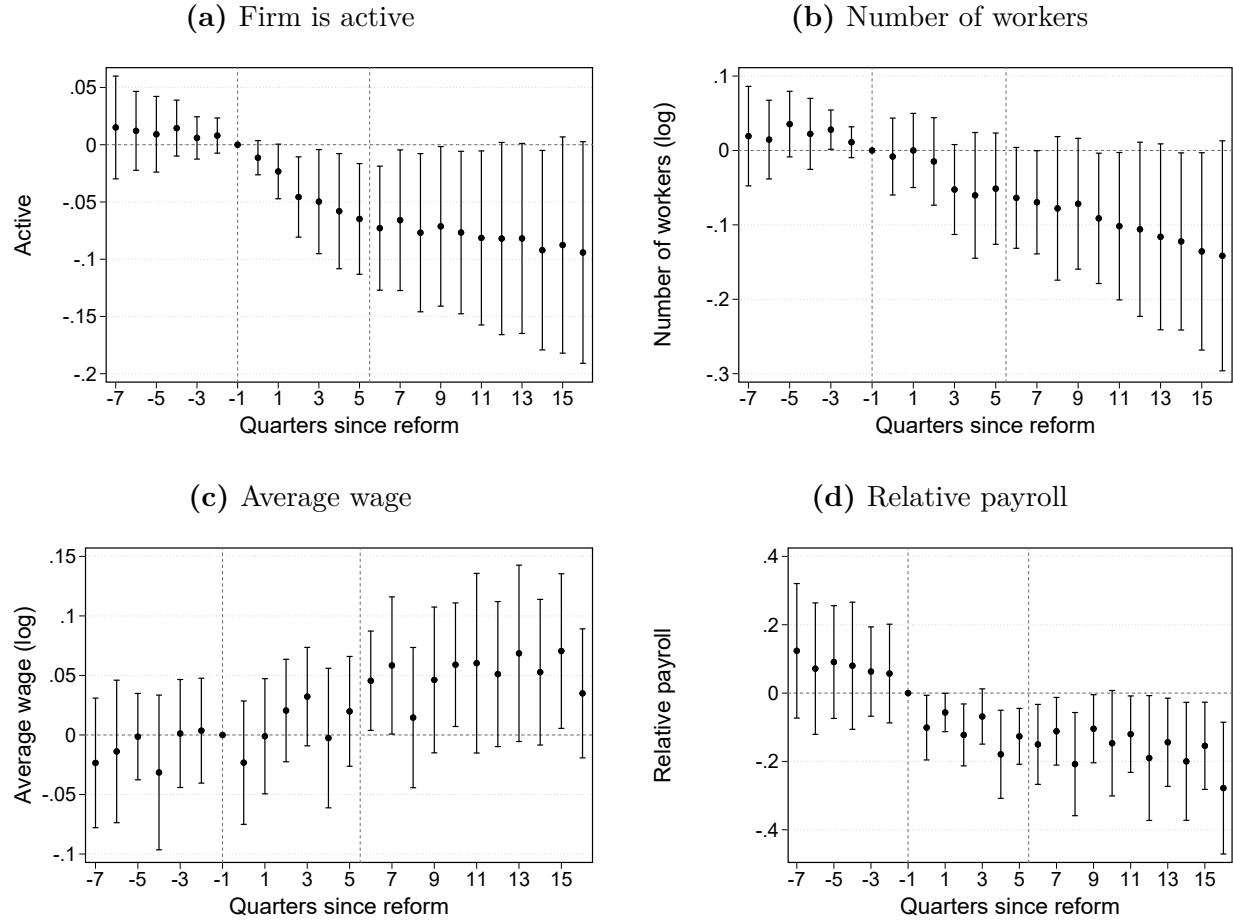
(b) Account ownership in Latin America



Notes: Panel (a) plots the share of adults aged 15 and older who own an account against log GDP per capita (current US dollars) in 2011. Panel (b) shows account ownership over time in Uruguay (red) and Argentina, Brazil, Costa Rica, Guatemala, Mexico, and Panama (blue). The vertical line marks 2015, when the reform began. Account ownership includes accounts at financial institutions or mobile-money providers. Sources: World Bank Global Findex Database for account ownership and World Development Indicators for GDP per capita.

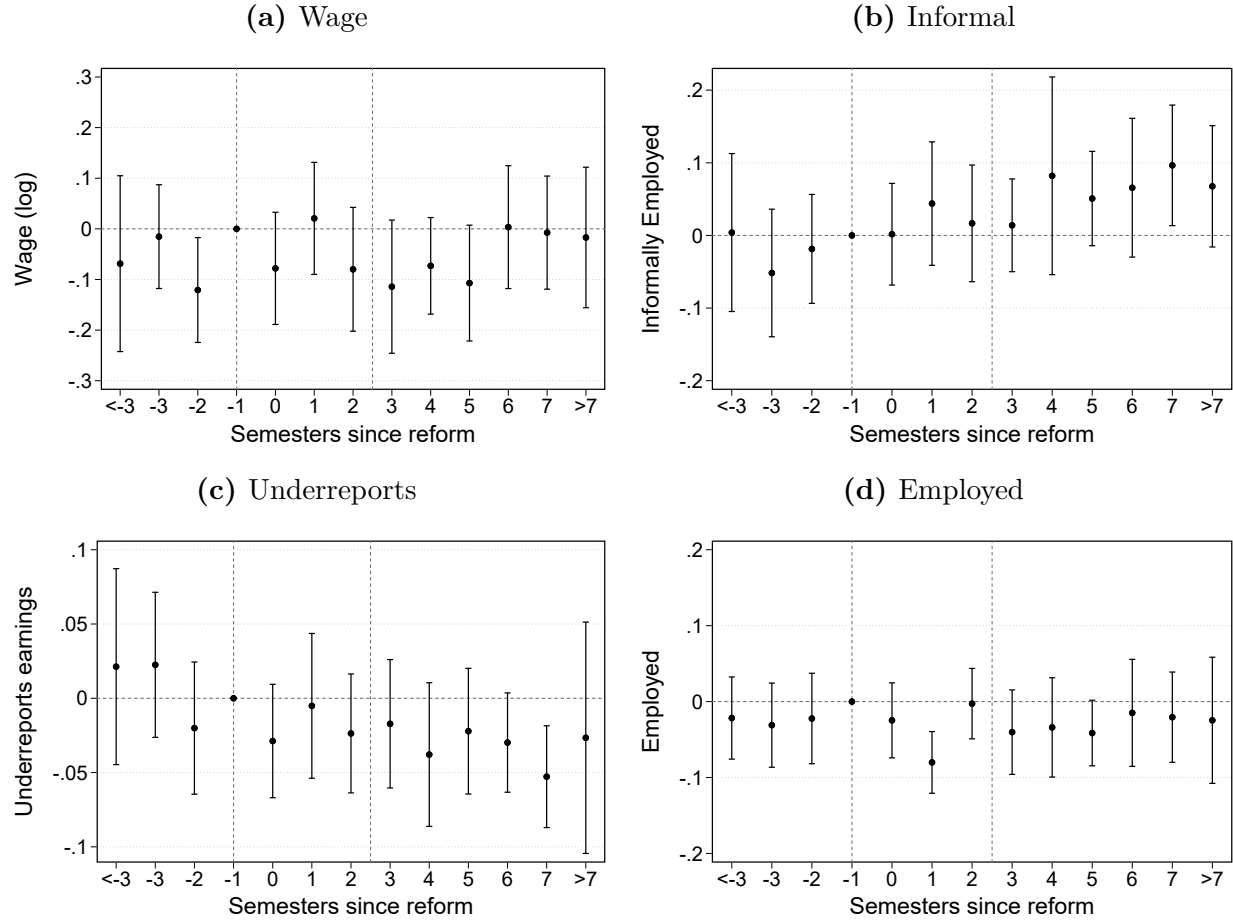
C. Omitting the retail and restaurant sectors

Figure C.I: Main results (administrative data) - omitting retail and restaurants



Notes: Baseline event studies excluding sectors 47 and 56 (retail and food services). Dashed lines mark the phase-in and mandate periods. Bars show 95% confidence intervals using sector-clustered standard errors.

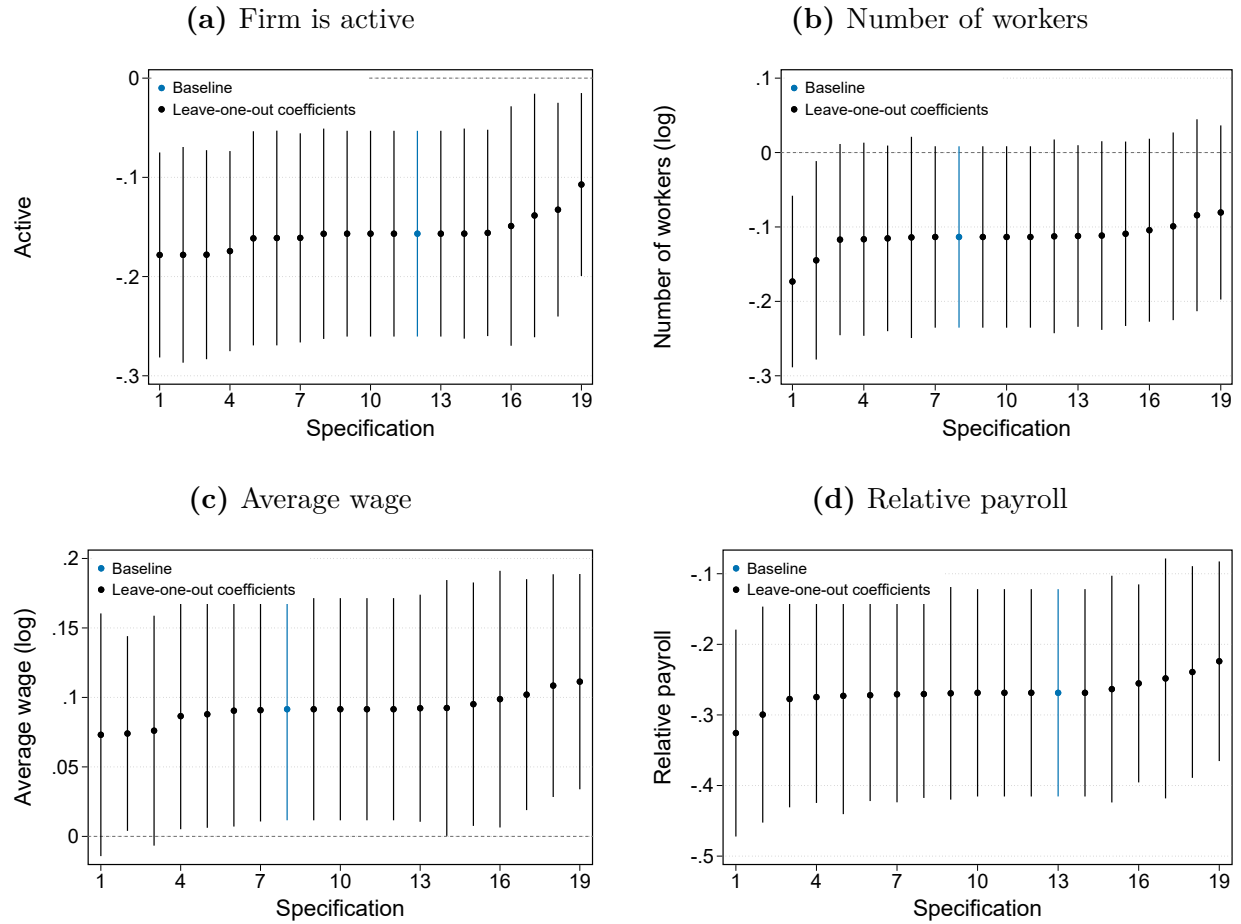
Figure C.II: Main results (survey data) - omitting retail and restaurants



Notes: This figure shows OLS estimates of the β_k coefficients from equation 7 grouped into six-month periods, omitting the retail and restaurant sectors. In panel (a) the dependent variable is the natural logarithm of the wage. In panel (b) the dependent variable is an indicator equal to 1 if the worker is informally employed. In panel (c) the dependent variable is an indicator equal to 1 if the worker underreports their labor earnings for tax purposes. In panel (d) the dependent variable is an indicator equal to 1 if the worker is employed and 0 otherwise. The first dashed line indicates the beginning of the phase-in period and the second dashed line indicates the beginning of the mandate period. Capped vertical bars represent 95% confidence intervals from standard errors clustered at the sector level.

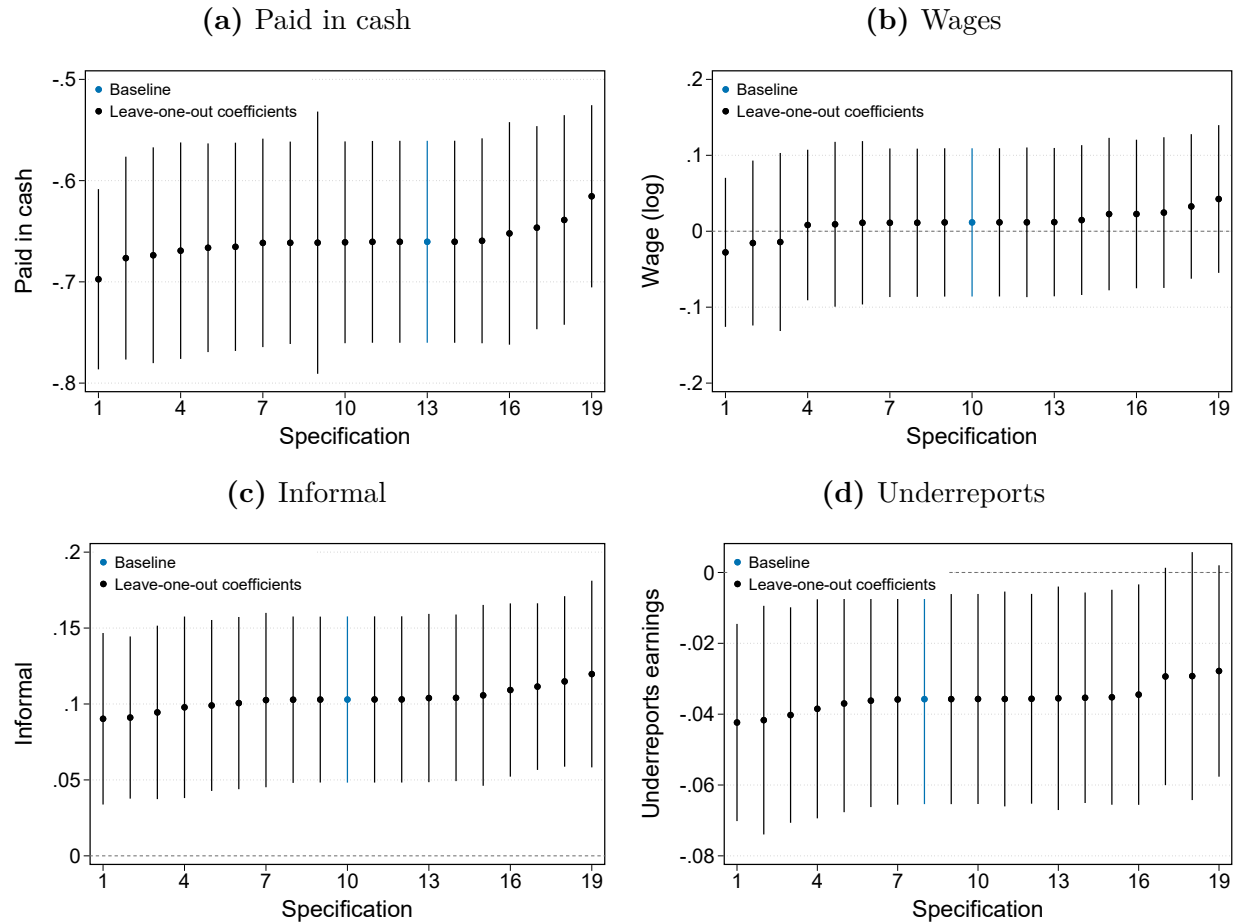
D. Leave-one-out robustness checks

Figure D.I: Leave-one-out robustness checks - Administrative data



Notes: Pooled long-run estimates for 2019Q2–2019Q4, dropping one broad industry at a time. Baseline estimates are blue; leave-one-out estimates are black. Specifications are sorted separately within each panel. Bars show 95% confidence intervals using sector-clustered standard errors.

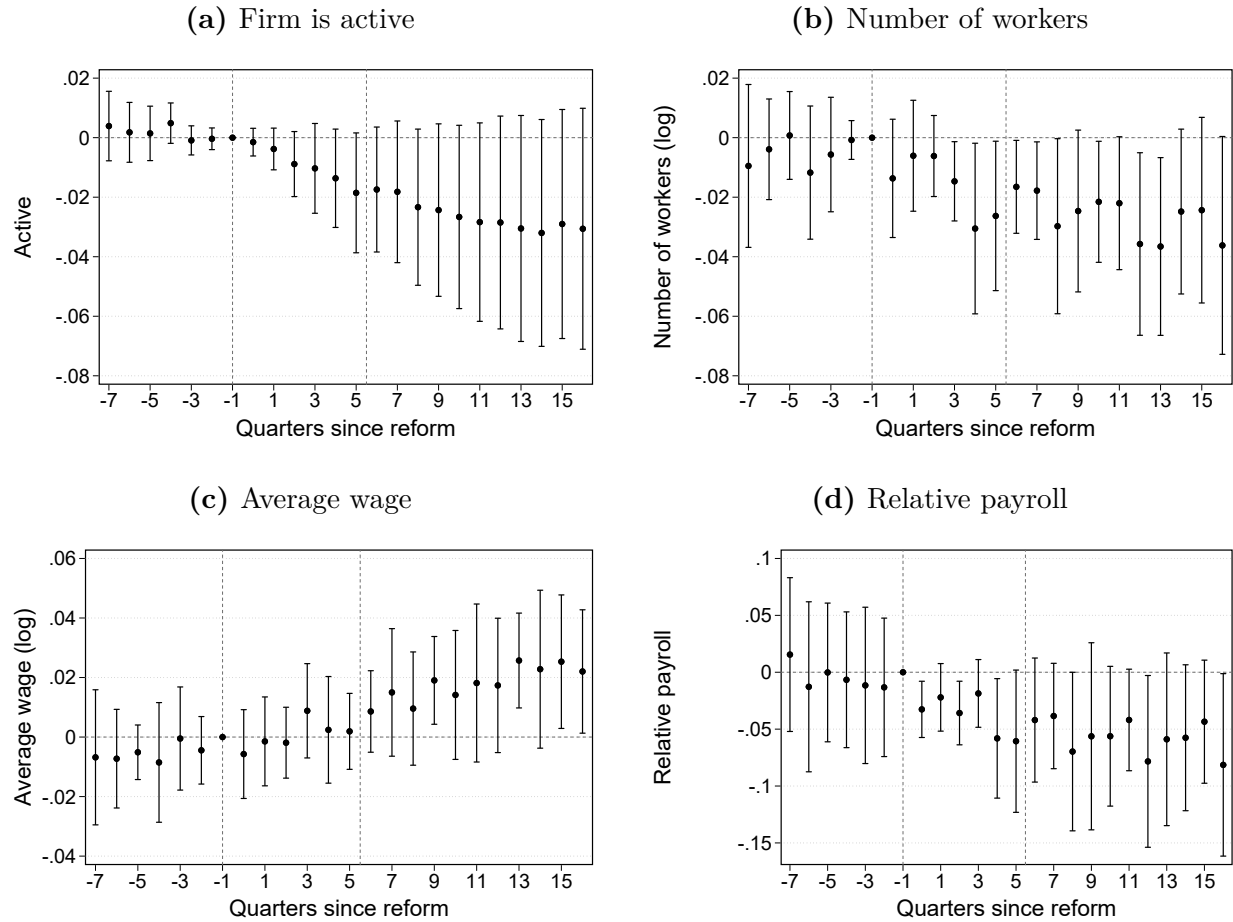
Figure D.II: Leave-one-out robustness checks - Survey data



Notes: Pooled long-run estimates for 2019Q2–2019Q4, dropping one broad industry at a time. Baseline estimates are blue; leave-one-out estimates are black. Specifications are sorted separately within each panel. Bars show 95% confidence intervals using sector-clustered standard errors.

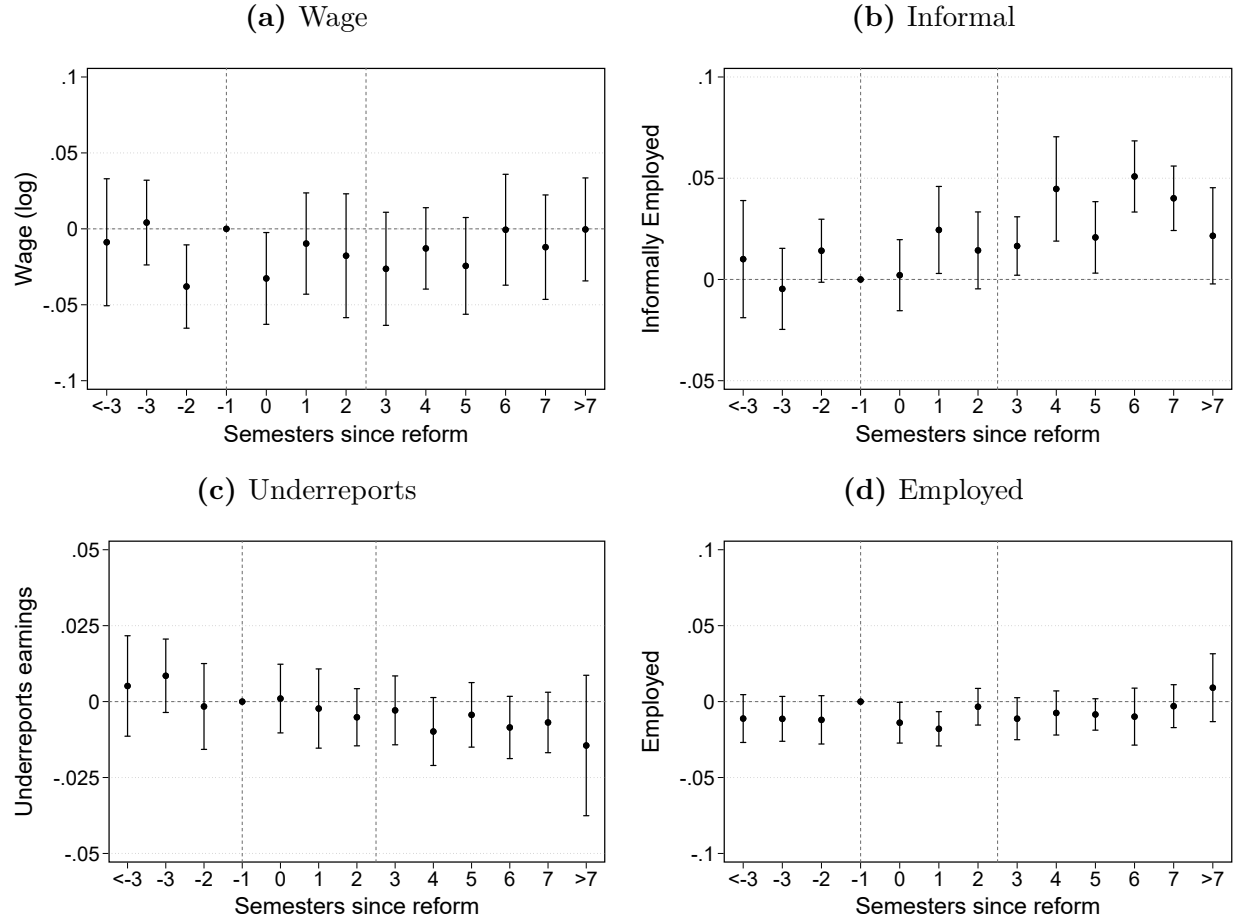
E. Discrete cash-intensiveness robustness check

Figure E.I: Main results (administrative data) - discrete cash intensity



Notes: Event studies replacing continuous exposure with an indicator for above-median pre-reform cash intensity within broad industries. Dashed lines mark the phase-in and mandate periods. Bars show 95% confidence intervals using sector-clustered standard errors.

Figure E.II: Main results (survey data) - discrete cash intensiveness robustness check



Notes: This figure shows OLS estimates of the β_k coefficients from equation 7 defining cash intensiveness discretely as above and below median pre-reform cash intensity grouped into six-month periods. In panel (a) the dependent variable is the natural logarithm of the wage. In panel (b) the dependent variable is an indicator equal to 1 if the worker is informally employed. In panel (c) the dependent variable is an indicator equal to 1 if the worker underreports their labor earnings for tax purposes. In panel (d) the dependent variable is an indicator equal to 1 if the worker is employed and 0 otherwise. The first dashed line indicates the beginning of the phase-in period and the second dashed line indicates the beginning of the mandate period. Capped vertical bars represent 95% confidence intervals from standard errors clustered at the sector level.

F. Derivations for conceptual framework

This appendix derives first-order conditions (FOCs), states sufficient regularity and interiority conditions, and derives sorting and comparative statics under explicit assumptions. We use the main-text notation and assume Inada conditions for $y = \theta f(L)$: $f' > 0$, $f'' < 0$, $\lim_{L \rightarrow 0^+} f'(L) = +\infty$, and $\lim_{L \rightarrow \infty} f'(L) = 0$. Formal firms' profit is

$$\pi_F(\theta; L, \gamma, \eta) = \theta f(L) - \left[1 + \tau\gamma + \phi\tau(1 - \gamma)p(\gamma, \eta, L) \right] wL - F, \quad (\text{F.1})$$

with $p_\gamma < 0$, $p_L > 0$, $p_\eta > 0$, and a constant penalty multiplier $\phi > 1$. Enforcement affects detection probability, not the payment conditional on detection. Informal firms' profit is

$$\pi_I(\theta; L) = \theta f(L) - w\kappa(L), \quad \kappa' > 0, \quad \kappa'' > 0. \quad (\text{F.2})$$

F.1 Optimal input choices (FOCs) and interiority

Informal firm. Maximizing (F.2) over $L \geq 0$ gives

$$\theta f'(L_I^*) = w\kappa'(L_I^*).$$

The left-hand side is strictly decreasing in L , with $\lim_{L \downarrow 0} \theta f'(L) = +\infty$ by the Inada condition. The right-hand side is strictly increasing in L . If $\kappa'(0) \in (0, \infty)$ and $\lim_{L \rightarrow \infty} \kappa'(L) = \infty$, the two sides cross exactly once, delivering a unique interior solution $L_I^* > 0$. Implicit differentiation gives

$$\frac{\partial L_I^*}{\partial \theta} = \frac{f'(L_I^*)}{w\kappa''(L_I^*) - \theta f''(L_I^*)} > 0.$$

Formal firm. Maximizing (F.1) over (L, γ) yields two FOCs.

FOC in γ . Holding L fixed,

$$\frac{\partial \pi_F}{\partial \gamma} = -wL \left\{ \tau + \phi\tau \left[-p(\gamma, \eta, L) + (1 - \gamma)p_\gamma(\gamma, \eta, L) \right] \right\} = 0, \quad (\text{F.3})$$

which holds if and only if

$$1 = \phi \left[p(\gamma, \eta, L) - (1 - \gamma)p_\gamma(\gamma, \eta, L) \right]. \quad (\text{F.4})$$

Let

$$g(\gamma; \eta, L) := \phi \left[p(\gamma, \eta, L) - (1 - \gamma)p_\gamma(\gamma, \eta, L) \right].$$

A sufficient condition for a unique interior solution $\gamma^*(\eta, L) \in (0, 1)$ to (F.4) is

$$g_\gamma(\gamma; \eta, L) = \phi \left[2p_\gamma(\gamma, \eta, L) - (1 - \gamma)p_{\gamma\gamma}(\gamma, \eta, L) \right] < 0 \quad \text{for all } \gamma \in [0, 1], \quad (\text{F.5})$$

together with the boundary conditions

$$g(0; \eta, L) > 1 \quad \text{and} \quad g(1; \eta, L) = \phi p(1, \eta, L) < 1. \quad (\text{F.6})$$

FOC in L. Differentiating (F.1) with respect to L gives

$$\frac{\partial \pi_F}{\partial L} = \theta f'(L) - w \left\{ 1 + \tau\gamma + \phi\tau(1 - \gamma)p(\gamma, \eta, L) \right\} - wL\phi\tau(1 - \gamma)p_L(\gamma, \eta, L) = 0, \quad (\text{F.7})$$

which holds if and only if

$$\theta f'(L_F^*) = w \left\{ 1 + \tau\gamma^* + \phi\tau(1 - \gamma^*) \left[p(\gamma^*, \eta, L_F^*) + L_F^* p_L(\gamma^*, \eta, L_F^*) \right] \right\}. \quad (\text{F.8})$$

Holding γ fixed, a sufficient second-order condition for the employment choice is

$$\frac{\partial^2 \pi_F}{\partial L^2} = \theta f''(L) - w\phi\tau(1 - \gamma) \left[2p_L(\gamma, \eta, L) + L p_{LL}(\gamma, \eta, L) \right] < 0,$$

which holds if $f'' < 0$, $p_L > 0$, and $p_{LL} \geq 0$. This is a conditional curvature condition, not by itself sufficient for joint concavity in (L, γ) . We additionally assume that a maximizer exists with $L_F^* > 0$, that (F.5)–(F.6) hold over the relevant employment range, and that formal profits are jointly strictly concave on the relevant convex choice set. A sufficient curvature restriction on its interior is

$$\pi_{LL} < 0, \quad \pi_{\gamma\gamma} < 0, \quad D := \pi_{LL}\pi_{\gamma\gamma} - \pi_{L\gamma}^2 > 0, \quad (\text{F.9})$$

where subscripts denote derivatives of π_F . These are additional regularity assumptions, not consequences of the signs of the own second derivatives alone. Together they deliver a unique interior joint optimum over this choice set.

Differentiating both FOCs at this optimum gives

$$\begin{pmatrix} \pi_{LL} & \pi_{L\gamma} \\ \pi_{L\gamma} & \pi_{\gamma\gamma} \end{pmatrix} \begin{pmatrix} dL_F^* \\ d\gamma^* \end{pmatrix} = - \begin{pmatrix} f'(L_F^*) \\ 0 \end{pmatrix} d\theta - \begin{pmatrix} \pi_{L\eta} \\ \pi_{\gamma\eta} \end{pmatrix} d\eta.$$

Consequently,

$$\frac{\partial L_F^*}{\partial \theta} = -\frac{\pi_{\gamma\gamma} f'(L_F^*)}{D} > 0, \quad \frac{\partial \gamma^*}{\partial \theta} = \frac{\pi_{L\gamma} f'(L_F^*)}{D}.$$

At the reporting FOC, $\pi_{L\gamma} = w\tau L_F^* g_L$, so the latter derivative is positive under the condition $g_L > 0$ stated below. The joint employment response to enforcement is

$$\frac{\partial L_F^*}{\partial \eta} = \frac{\pi_{L\gamma}\pi_{\gamma\eta} - \pi_{\gamma\gamma}\pi_{L\eta}}{D},$$

whose sign is not fixed by joint concavity alone.

F.2 Comparative statics for reporting

Totally differentiating $g(\gamma^*; \eta, L_F^*) = 1$ gives

$$g_\gamma d\gamma^* + g_\eta d\eta + g_L dL_F^* = 0,$$

where

$$g_\eta = \phi \left[p_\eta - (1 - \gamma^*) p_{\gamma\eta} \right]$$

and

$$g_L = \phi \left[p_L - (1 - \gamma^*) p_{\gamma L} \right].$$

All terms involving p and its derivatives are evaluated at (γ^*, η, L_F^*) .

Under (F.5), $g_\gamma < 0$. If $p_\eta > 0$ and $p_{\gamma\eta} \leq 0$, then $g_\eta > 0$. Therefore, holding scale fixed,

$$\left. \frac{\partial \gamma^*}{\partial \eta} \right|_L = -\frac{g_\eta}{g_\gamma} > 0.$$

Stronger enforcement thus increases the reported share of wages holding employment fixed.

If, in addition, $p_L > 0$ and $p_{\gamma L} \leq 0$, then $g_L > 0$ and

$$\left. \frac{\partial \gamma^*}{\partial L} \right|_\eta = -\frac{g_L}{g_\gamma} > 0.$$

Because more productive formal firms choose a larger scale, this implies

$$\frac{\partial \gamma^*}{\partial \theta} > 0.$$

Allowing formal employment to respond to enforcement, the total effect is

$$\frac{d\gamma^*}{d\eta} = -\frac{g_\eta + g_L \frac{dL_F^*}{d\eta}}{g_\gamma}.$$

Thus,

$$\frac{d\gamma^*}{d\eta} > 0$$

provided that $g_\eta + g_L dL_F^*/d\eta > 0$: any enforcement-induced decline in formal employment must not be sufficiently large to offset the direct enforcement effect. The employment derivative is determined by the joint FOC system above; this reporting condition is an additional restriction on that response.

F.3 Sorting: existence and uniqueness of a productivity cutoff

Let

$$\Pi_F(\theta; \eta) = \pi_F(\theta; L_F^*(\theta, \eta), \gamma^*(\theta, \eta), \eta),$$

$$\Pi_I(\theta) = \pi_I(\theta; L_I^*(\theta)),$$

and

$$\Delta(\theta; \eta) = \Pi_F(\theta; \eta) - \Pi_I(\theta).$$

A firm operates formally if and only if

$$\Delta(\theta; \eta) \geq 0.$$

By the envelope theorem,

$$\frac{\partial \Pi_F(\theta; \eta)}{\partial \theta} = f(L_F^*(\theta, \eta)),$$

$$\frac{\partial \Pi_I(\theta)}{\partial \theta} = f(L_I^*(\theta)),$$

and therefore

$$\frac{\partial \Delta(\theta; \eta)}{\partial \theta} = f(L_F^*(\theta, \eta)) - f(L_I^*(\theta)). \quad (\text{F.10})$$

We impose the single-crossing condition

$$\frac{\partial \Delta(\theta; \eta)}{\partial \theta} > 0 \quad \text{over the relevant range of } \theta. \quad (\text{F.11})$$

A sufficient condition is that formal firms operate at a larger optimal scale than informal firms around and above the formality cutoff:

$$L_F^*(\theta, \eta) > L_I^*(\theta).$$

We also assume that formal operation is unprofitable for sufficiently low-productivity firms because of the fixed cost F , but profitable for sufficiently high-productivity firms. Thus, for some $\underline{\theta} < \bar{\theta}$,

$$\Delta(\underline{\theta}; \eta) < 0 \quad \text{and} \quad \Delta(\bar{\theta}; \eta) > 0.$$

Continuity and the single-crossing condition imply the existence of a unique productivity cutoff $\theta^*(\eta)$ satisfying

$$\Delta(\theta^*(\eta); \eta) = 0. \tag{F.12}$$

Consequently,

$$\theta < \theta^*(\eta) \quad \Rightarrow \quad \Pi_I(\theta) > \Pi_F(\theta; \eta),$$

while

$$\theta > \theta^*(\eta) \quad \Rightarrow \quad \Pi_F(\theta; \eta) > \Pi_I(\theta).$$

F.4 Comparative statics in η

By the envelope theorem, the direct effect of enforcement on optimized formal profits is

$$\frac{\partial \Pi_F(\theta; \eta)}{\partial \eta} = -wL_F^* \tau (1 - \gamma^*) \phi p_\eta(\gamma^*, \eta, L_F^*) < 0. \tag{F.13}$$

Informal profits do not depend directly on η , so

$$\frac{\partial \Delta(\theta; \eta)}{\partial \eta} < 0.$$

Implicitly differentiating (F.12) gives

$$\frac{d\theta^*(\eta)}{d\eta} = - \frac{\frac{\partial \Delta(\theta^*(\eta); \eta)}{\partial \eta}}{\frac{\partial \Delta(\theta^*(\eta); \eta)}{\partial \theta}} > 0. \tag{F.14}$$

Stronger enforcement therefore raises the productivity cutoff for formal operation. Some relatively low-productivity firms move from formality into informality. Under the reporting-response condition in Appendix F.2, firms that remain formal report a larger share of their wages.

Aggregate formal employment. Normalize the mass of firms to one and let $G(\theta)$ be a fixed productivity distribution with density $\rho(\theta)$. Aggregate formal employment is

$$E_F(\eta) = \int_{\theta^*(\eta)}^{\infty} L_F^*(\theta, \eta) dG(\theta).$$

Assuming the integral is finite and differentiation under the integral is valid,

$$\frac{dE_F}{d\eta} = -L_F^*(\theta^*, \eta)\rho(\theta^*)\frac{d\theta^*}{d\eta} + \int_{\theta^*}^{\infty} \frac{\partial L_F^*(\theta, \eta)}{\partial \eta} dG(\theta). \quad (\text{F.15})$$

The first term is negative when the density at the cutoff is positive: firms leaving formality reduce formal employment. The second term captures employment adjustments at continuing formal firms and need not be negative. Aggregate formal employment falls if those adjustments do not offset the extensive-margin loss. Nonincreasing employment at every continuing formal firm is sufficient, but is not implied by the cutoff result alone.

F.5 Implications for payroll-tax revenue

Let $E_F(\eta)$ denote aggregate formal employment and let $\tilde{w}_F(\eta)$ denote the average reported wage tax base per formal worker. Payroll-tax revenue is

$$R(\eta) = \tau E_F(\eta) \tilde{w}_F(\eta).$$

Differentiating gives

$$\frac{dR(\eta)}{d\eta} = \tau \left[\frac{dE_F(\eta)}{d\eta} \tilde{w}_F(\eta) + E_F(\eta) \frac{d\tilde{w}_F(\eta)}{d\eta} \right].$$

Under the reporting-response condition, improved reporting within continuing formal firms tends to raise revenue, while firms leaving formality remove their reported payroll from the tax base. However, the sign of $dE_F/d\eta$ also depends on employment adjustments at continuing firms, as shown in (F.15). Moreover, $\tilde{w}_F$ is an employment-weighted average: its response depends on reporting within firms as well as changes in firm composition and employment weights. We therefore do not infer either aggregate derivative solely from the cutoff or within-firm reporting result. The net revenue effect is not signed without further restrictions on these responses.

Remark. The strict comparative statics above concern an interior, regular optimum. If the sufficient conditions in (F.5)–(F.6) or (F.9) fail, corner solutions or multiple optima may arise, and the interior derivatives do not automatically extend to these cases. Continuity, single crossing, and the stated boundary conditions can still deliver a unique formality cutoff, but its response to enforcement requires a separate argument. In particular, a fully compliant firm ($\gamma^* = 1$) has no direct enforcement cost in this specification, so a strictly rising cutoff is not guaranteed at that corner.

G. Accounting decompositions

This appendix defines the outcomes used in Panel B of Table 1. All components are constructed at the firm–quarter level. Within each decomposition, we use a common sample and identical regressors and fixed effects; linearity therefore ensures that the component coefficients sum to the coefficient for the total. These are accounting contributions, not separately identified causal mechanisms.

G.1 Quarterly employment growth

Let L_{it} be firm i 's recorded employment in quarter t . Hires H_{it} are workers present at t but absent at $t - 1$; separations D_{it} are workers present at $t - 1$ but absent at t . Thus $L_{it} - L_{i,t-1} = H_{it} - D_{it}$. For firms active throughout the sample, define

$$a_{it} = \begin{cases} \frac{\log L_{it} - \log L_{i,t-1}}{L_{it} - L_{i,t-1}}, & L_{it} \neq L_{i,t-1}, \\ \frac{1}{L_{i,t-1}}, & L_{it} = L_{i,t-1}. \end{cases} \quad (\text{G.1})$$

The second case is the continuous limit. The exact decomposition is

$$\Delta \log L_{it} = \underbrace{a_{it} H_{it}}_{\text{Hiring component}} + \underbrace{(-a_{it} D_{it})}_{\text{Separation component}}. \quad (\text{G.2})$$

When employment is unchanged, hiring and separation contributions cancel, even if both flows are positive. A positive coefficient on the signed separation component indicates a less negative contribution from separations. These outcomes describe quarterly growth, not cumulative employment changes relative to 2015Q3.

G.2 Average reported wages

Let 0 denote 2015Q3, W_{it} the set of workers recorded at firm i in quarter t , and $S_{it} = W_{i0} \cap W_{it}$ the workers shared between the two dates. Write $\bar{w}_{it}$ for the average real reported wage of all current workers and $\bar{w}_{iu}^{S_{it}}$ for the average wage of this shared group at date $u \in \{0, t\}$. Then

$$\log \frac{\bar{w}_{it}}{\bar{w}_{i0}} = \underbrace{\log \frac{\bar{w}_{it}^{S_{it}}}{\bar{w}_{i0}^{S_{it}}}}_{\text{Stayer wage component}} + \underbrace{\log \frac{\bar{w}_{i0}^{S_{it}}}{\bar{w}_{i0}}}_{\text{Departure component}} + \underbrace{\log \frac{\bar{w}_{it}}{\bar{w}_{it}^{S_{it}}}}_{\text{Entry component}}. \quad (\text{G.3})$$

The departure component is positive when the shared workers earned more at baseline than the full baseline workforce. The entry component is positive when entrants raise the current mean above that of shared workers. Membership in S_{it} varies with t and does not require uninterrupted employment between the two dates. We retain always-active firms with nonempty overlap and positive, observed group means in every quarter. Individual zero wages enter the means; absent workers do not.

G.3 Relative payroll

Let P_{it} be real payroll and $B_i > 0$ the firm's mean pre-reform payroll used to normalize the main payroll outcome. Unlike the wage decomposition, this exercise retains inactive firm-quarters with zero payroll. Using the same baseline and shared-worker set as above, define

$$\begin{aligned} P_{it}^S &= \sum_{k \in S_{it}} w_{kit}, & P_{i0,t}^S &= \sum_{k \in S_{it}} w_{ki0}, \\ P_{it}^D &= \sum_{k \in W_{i0} \setminus W_{it}} w_{ki0}, & P_{it}^E &= \sum_{k \in W_{it} \setminus W_{i0}} w_{kit}. \end{aligned}$$

These are current and baseline payroll for shared workers, baseline payroll for departed workers, and current payroll for entrants, respectively. Empty groups contribute zero. The unadjusted identity is

$$\frac{P_{it} - P_{i0}}{B_i} = \underbrace{\frac{P_{it}^S - P_{i0,t}^S}{B_i}}_{\text{Stayer wage component}} + \underbrace{\frac{-P_{it}^D}{B_i}}_{\text{Departure component}} + \underbrace{\frac{P_{it}^E}{B_i}}_{\text{Entry component}}. \quad (\text{G.4})$$

To match the winsorized payroll outcome, let R_{it}^c denote the ratio used in the main regression and set $q_{it} = R_{it}^c / (P_{it}/B_i)$ when payroll is positive, and $q_{it} = 1$ otherwise. We proportionally scale current payments to shared workers and entrants, leaving baseline payments unchanged. The reported components therefore satisfy

$$R_{it}^c - \frac{P_{i0}}{B_i} = \frac{q_{it}P_{it}^S - P_{i0,t}^S}{B_i} - \frac{P_{it}^D}{B_i} + \frac{q_{it}P_{it}^E}{B_i}. \quad (\text{G.5})$$

Subtracting P_{i0}/B_i , which is constant within firm, does not change the exposure coefficients in a regression with firm fixed effects. Hence these components sum to the main relative-payroll coefficient. For an inactive firm, the entire payroll loss is assigned to departures. Entry and departure refer to administrative payroll records and need not represent the start or end of an underlying employment relationship.